Economic Forecasting with an Agent-Based Model of the United States:
Calibration, Diagnosis, and Out-of-Sample Performance

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Abstract

We port the agent-based macroeconomic model of Poledna et al. [2023] to the United States and subject it to that paper's own out-of-sample protocol: 61 quarterly forecast origins from 2010Q2 to 2025Q2, horizons of 1, 2, 4, 8 and 12 quarters, 500 free-running stochastic paths per origin, and ten statistical benchmarks scored on identical cells. The first pass produced a defect rather than a result: real GDP growth was under-forecast by 1.3 to 2.9 percentage points at every horizon. We trace it by matched-seed experiment to the calibration artifact's opening commodity balance, which does not clear sector by sector — modelled uses differ from supply by −70.6% to +47.6% across 68 commodities — colliding with a firm capacity envelope frozen at $K_i \kappa_i = Y_i / 0.85$ because investment is replacement-only. Over-demanded sectors sit permanently against their ceiling, under-demanded ones permanently idle, and the one-sided production constraint prevents the idle capacity from compensating. This accounts for 76 % of the bias; a growth expectation estimated by ordinary least squares AR(1) on the *log level* of an I(1)-with-drift series, which delivers only 32–63 % of the in-sample trend, accounts for the remainder. We repair both on accounting grounds alone — biproportional (RAS) reconciliation of the use side that retains measured BEA imports, and a random-walk-with-drift expectation — with neither repair *selected* using forecast errors, and re-score on identical cells with identical seeds. The exercise remains explicitly mixed-vintage: 2024 structure and current-vintage data are used at historical origins. The repaired model attains the best weighted RMSE ratio of all fourteen scored columns on {real GDP growth, GDP-deflator inflation} in all three sample tracks (0.830, 0.829 and 0.796 against a VAR(1) anchor); real-GDP bias falls below 0.21 percentage points at every horizon and empirical 90 % interval coverage moves from 0.654 to 0.929. The GDP deflator is unchanged, and bit-identical at $h = 1$, which is the signature the diagnosis predicted. Two measured defects bound every number: capital never accumulates, so growth drains a fixed 17.6 % capacity headroom and the result is validated only to $h \leq 12$; and unemployment collapses to about 1 %, is excluded from every weighted score, and became worse. This is a revised-data, mixed-vintage diagnostic, not a real-time forecasting claim.

Keywords: agent-based macroeconomics; out-of-sample forecast evaluation; input–output reconciliation; supply–use balance; model diagnosis.

Reader's note. This paper is written for a technically trained reader who is not an economist by training. Economic machinery is defined at first use, and the accounting identities that carry the argument are written out rather than assumed. Every number is traceable to a committed artifact; Appendix A gives the paths, hashes and commands.

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1 Introduction

1.1 The Poledna programme

Macroeconomic forecasting is dominated by two model classes. *Time-series* models — vector autoregressions (VARs) and their Bayesian variants — treat the economy as a low-dimensional stochastic process and extrapolate its estimated dynamics; they are accurate at short horizons and structurally mute. *Dynamic stochastic general equilibrium* (DSGE) models [Smets and Wouters, 2007] write down a small number of optimising representative agents, impose market clearing at every date, and estimate the resulting cross-equation restrictions; they are structurally interpretable and, at forecast horizons, rarely better than a well-specified VAR.

Agent-based models (ABMs) take a third route [Farmer and Foley, 2009]. They simulate a population of heterogeneous entities — firms, households, a bank, a government — each following explicit behavioural rules, and let aggregates emerge from decentralised interaction. Markets do not clear by assumption; they clear, or fail to clear, through search and matching, with rationing as an observable outcome. For two decades this class was excluded from the forecasting conversation on the grounds that it was unfalsifiable in the only way that matters: nobody had run one against the standard benchmarks on a rolling out-of-sample grid.

Poledna et al. [2023] closed that gap. They calibrated a large ABM of Austria one-to-one against the national accounts, ran free-running forecasts from quarterly origins over 2010Q2–2016Q4 at horizons $h \in \{1, 2, 4, 8, 12\}$ with 500 stochastic paths each, and scored the ensemble mean against AR, VAR and Bayesian DSGE benchmarks. The ABM was competitive, and better than the DSGE at most horizons. That result is the reference point of the present work, and its protocol is the one reproduced here.

1.2 What this paper does

This paper reports a port of that programme to the United States, and it reports what the port found. The arc is: *built → scored → found defective → diagnosed → repaired on accounting grounds → re-scored on identical cells.* That arc, and not the rank it terminates in, is the contribution.

1. **Port and first-pass validation** (§3–§6). The BeforeIT implementation [Glielmo et al., 2025] of the Poledna model is calibrated against the U.S. Bureau of Economic Analysis (BEA) supply–use system and run over 61 quarterly origins, 2010Q2–2025Q2, at $h \in \{1, 2, 4, 8, 12\}$ with 500 paths per origin and zero path failures, against ten statistical benchmarks on matched cells.
2. **Defect discovery** (§7). The first-pass model (`beforeit_abm_us_v1`) is competitive on the full sample and eleventh of fourteen once the pandemic cells are removed, and it under-forecasts real GDP growth by 1.3 to 2.9 percentage points at every horizon.
3. **Diagnosis** (§8). Matched-seed experiments attribute 76 % of the bias to a non-clearing opening commodity balance colliding with a frozen capacity envelope, and 24 % to a mis-specified growth-expectation estimator, with an explicit falsification table for the competing hypotheses.
4. **Structural repair** (§9). Both defects are corrected in the calibration artifact using past-only information and no reference to any forecast error: a biproportional (RAS) reconciliation of the use side that retains the measured import levels, and a random-walk-with-drift growth expectation.
5. **Re-validation** (§10). The repaired model (`beforeit_abm_us_v2`) is scored jointly with the unrepaired one, on identical common cells with identical random seeds, in a single run.

1.3 What the reader should take away, and what they should not

The repaired model ranks first of fourteen on the headline target pair in all three sample tracks. That is a real result on a matched grid, and it is also a *revised-data, mixed-vintage diagnostic*: the data are current-vintage throughout, and the input–output structure is frozen at 2024 and carried backwards to every historical origin. No real-time information set was reconstructed, no DSGE benchmark is in the field, and no formal forecast-comparison inference was performed, so adjacent ranks are not statistically distinguishable. Two open defects — capital that never accumulates and a labour block that overheats after the repair — bound the validity of every number reported here. These limits are stated in §12 and are not incidental to the result; they are part of it.

2 The model

This section is a compact overview sufficient to read the results. The full equation system is in Poledna et al. [2023] and its implementation in Glielmo et al. [2025]; nothing in the behavioural block was modified for this work except the one flag-gated expectation change documented in §9.

2.1 Agents and the flow of a quarter

The economy contains six classes of agent.

Firms.

$\sum_s I_s$ firms distributed over S industries. Each firm i in industry s produces a single commodity with a *Leontief* technology [Leontief, 1936]: output requires labour, intermediate inputs and capital services in fixed proportions, with no substitution between them. Concretely, output is capped by the binding factor,

$$Y_i = \min\Big(\underbrace{\alpha_s N_i}_{\text{labour}},\ \underbrace{\beta_s M_i}_{\text{materials}},\ \underbrace{\kappa_s K_i}_{\text{capital}}\Big), \tag{1}$$

where N_i is employment, M_i the intermediate-input bundle, K_i the capital stock, and $(\alpha_s, \beta_s, \kappa_s)$ are industry productivities read off the input–output table. The third term is the *capacity envelope*; it is the object at the centre of §8.

Households.

Worker households supply labour and consume; inactive households receive transfers and consume; firm owners receive dividends. Consumption follows a propensity rule on disposable income and liquid wealth, allocated across commodities by a fixed basket b_g^{HH}.

A bank.

One consolidated commercial bank intermediates deposits and loans subject to a capital-adequacy constraint, and rations credit when the constraint binds.

A central bank.

Sets the policy rate by a Taylor rule — a feedback rule raising the rate when inflation or an activity measure exceeds target — with inertia ρ, an intercept r^*, and response coefficients ξ_π and ξ_γ.

A government.

Collects taxes, pays wages, transfers and unemployment benefits, consumes a commodity basket c_g^G, and issues debt.

Table 1: The five model relations that the diagnosis of §8 turns on. Notation follows Poledna et al. [2023]; δ_s is the depreciation rate, ω the assumed normal capacity utilisation, Q_i^d desired sales and Q_i^s target production.

Relation	Form and role
Production, Leontief with a capacity cap	$Y_i = \min(\alpha_s N_i,\ \beta_s M_i,\ \kappa_s K_i)$. The min is *one-sided*: a firm below its cap cannot lend spare capacity to a firm above it.
Target output	$Q_i^s = Q_i^d\,(1+\gamma_e)$, where γ_e is the economy-wide expected growth rate.
Growth expectation (v1)	$\gamma_e = \exp(\hat{\alpha}\log Y_{\text{last}} + \hat{\beta})/Y_{\text{last}} - 1$ with $(\hat{\alpha}, \hat{\beta})$ from an OLS AR(1) with constant on the *log level* of the model's own aggregate output history.
Desired investment	$I_i^d = (\delta_i/\kappa_i)\min(Q_i^s,\ K_i\kappa_i)$, and $K' = K - (\delta/\kappa)Y + I$. When firms obtain their desired investment and produce at target these give $K' = K$ exactly: investment is *replacement-only.*
Opening capacity	$\kappa_s = \tau\,(\text{industry output})/(\text{fixed assets})/\omega$ and $K_i = Y_i/(\omega\kappa_i)$, so $K_i\kappa_i = Y_i/\omega$ identically. With $\omega = 0.85$ every firm opens at 85 % utilisation with $1/0.85 - 1 = 17.6$ % headroom.

A rest-of-world block.
Exports absorb a basket c_g^E; imports supply c_g^I. In this calibration the block is exogenous and is driven by estimated univariate processes.

Within a quarter, firms form expectations, choose target output and desired inputs, post vacancies and demand credit; the labour, credit and goods markets then clear through decentralised *search and matching* — buyers meet a random subset of sellers, transact at posted prices, and any unmet demand is simply not satisfied. Rationing is therefore a first-class observable of the model, reported here as *fill rates.* Insolvent firms exit and are replaced.

2.2 The behavioural core in five equations

Table 1 lists the equations that carry the argument of this paper. The remaining several dozen — price and wage setting, credit allocation, the bank's balance sheet, the fiscal block — are unchanged from the reference implementation and are not reproduced.

Two properties of Table 1 interact destructively, and §8 is about that interaction. First, the capacity envelope $K_i\kappa_i$ is fixed at construction in proportion to the *opening output composition.* Second, because investment is replacement-only, it never moves: measured over every experimental variant run for this paper, $K_{\text{end}}/K_1 = 1.0000$–$1.0009$. The capacity vector is therefore a constant, and the demand vector it must serve is a different constant, determined by the technology and final-demand coefficient matrices $(a_{sg}, b_g^{HH}, b_g^{CF}, c_g^G, c_g^E)$. If the two are inconsistent, nothing in the model can reconcile them.

The replacement-only investment rule is inherited from the upstream implementation, not introduced by this fork; the $\min(\cdot)$ in the desired-investment equation was already present in the upstream commit that this fork descends from.

3 United States calibration and data

3.1 The calibration artifact

A *calibration artifact* is a single serialised object holding every parameter and initial condition the simulator needs: the technology matrix, final-demand baskets, agent counts, tax and benefit rates, opening stocks, and the quarterly exogenous series. It is built once from public data and then loaded read-only by the forecasting harness.

Table 2: United States calibration at scale = 10^{-5}. One modelled firm stands for 10^5 real ones; the reference Austrian exercise ran at 1:1.

Commodities G	68	Economically active households H^{act}	1 826
Industries S (from 71 BEA industries)	68	Inactive households H^{inact}	1 005
Firms $\sum_s I_s$	130	Total population represented	$\approx$ 2 831
Government entities J	33	Foreign consumers L	65
Runtime: 0.085 s per 12-quarter path after compilation, single-threaded.			

The classification comes from the BEA summary-level supply–use system [U.S. Bureau of Economic Analysis, 2025a]. That source carries 68 observed commodities and 71 industries; the four retail industry columns (`441`, `445`, `452`, `4A0`) are summed into a single model sector `4A0`, giving 68 modelled industries matched one-to-one against the 68 commodities. Table 2 gives the resulting dimensions.

3.2 United States–specific choices in the artifact

Four choices depart from the reference specification and matter for what follows.

Explicit measured trade (`use_explicit_trade = true`).
Commodity imports are taken from the measured BEA T262 import matrix rather than being solved as the supply–use residual. This preserves the measured import level and is closer to the reference paper's stated intent of reading trade off the input–output table — but it leaves the sector-level balance unreconciled. This is the single most consequential U.S.-specific choice in the artifact and it is the subject of §8. The repair *retains* it.

Product-tax netting (`use_product_tax_netting = true`).
A valuation bridge, discussed in §4.

Discrepancy-derived opening inventories.
In the first-pass artifact, `use_commodity_balance_inventory = true` derives opening sector inventories S_s from the positive part of the negated statistical discrepancy. The pipeline source sets this flag to `false`; the artifact and the pipeline disagreed, and the artifact won for every origin the harness built. The repaired artifact sets it `false`, removing both the accuracy defect and the reproducibility hazard.

Policy rate on a quarterly basis.
The legacy field named `euribor` holds the U.S. policy rate, stored as $r = (\text{annual rate} + 1)^{1/4} - 1$. Comparison against the annualised-percentage-point effective federal funds rate (EFFR) therefore requires $100\,((1+r)^4 - 1)$. At 2024Q4 this gives 4.571 % against a panel value of 4.65 %.

3.3 The no-fitting firewall

That behavioural parameters have *not* been fitted to forecast errors was verified rather than assumed. The stored baseline carries firewall metadata recording that the parameters are raw, that no forecast-error-fitted parameters are present (`forecast_error_fitted_parameters = false`) and that no post-processing is embedded, with four restored parameters and seven removed output corrections. Numerically, at 2024Q4 the four Taylor-rule parameters take the values in Table 3; all $\approx$ 70 parameter keys agree between the stored baseline and a fresh derivation, neither equals the discarded forecast-fitted overrides, and the two produce bit-identical 12-quarter paths under the same seed.

Table 3: Raw-dynamics firewall check at 2024Q4. The right-hand column is the discarded forecast-error-fitted override layer, retained only for contrast; no result in this paper reads it.

Parameter	Stored baseline	Fresh derivation	Discarded fitted override
ρ	0.954811254910	0.954811254910	0.911458333333
r^*	−0.001043926436	−0.001043926436	−0.002333333333
ξ_π	0.789655283206	0.789655283206	0.708333333333
ξ_γ	0.532167779487	0.532167779487	0.258730279487

The firewall applies with equal force to the repair of §9. Both changes restore an accounting identity or correct an estimator that is wrong on its own terms, using past-only information. Re-estimating ω, the Taylor-rule coefficients or the expectation parameters against realised GDP paths would be behavioural tuning; it is named here explicitly as prohibited, and it was not done.

3.4 Absence of behavioural look-ahead

The exogenous arrays for government consumption, exports and imports extend up to twelve quarters past each origin in the artifact. This is behaviourally empty: only the element at the origin index is ever read, once, as a scalar, at construction, after which the series evolve by their own estimated processes. The claim was tested by mutation. Three runs at 2024Q4 under an identical seed — untruncated; truncated to the origin; and untruncated with *every post-origin entry overwritten by* -10^{99} — produced bit-identical real GDP, nominal GDP and policy-rate paths, with the poisoned run finite throughout. Truncation is applied anyway as past-only hygiene, and every estimation window terminates at the origin.

4 From Austria to the United States: what had to change, and why it was hard

The reference exercise is Austrian. Porting it is often described as a data-substitution task. It is not, and the difference is the origin of the defect diagnosed in §8.

4.1 A side-by-side comparison

4.2 Why each difference is hard, not merely different

Valuation bases. A supply–use system can be stated at *basic prices* (what the producer receives) or at *purchasers' prices* (what the buyer pays, including trade and transport margins and net product taxes). The model's coefficients are defined on the basic-price convention; the BEA tables are published at purchasers' prices. Bridging requires netting product taxes and reallocating margins, and every bridge leaves a residual somewhere. Eurostat hands the Austrian calibration a table that is already on the required basis; the U.S. calibration has to construct one.

The industry/commodity axis. Eurostat publishes symmetric product-by-product tables: rows and columns index the same 64 objects. BEA publishes a *make* table (which industry produces which commodity) and a *use* table (which industry consumes which commodity) over non-identical axes — 71 industries against 68 commodities. Projecting one onto the other is a modelling choice, not a lookup, and it is where an industry-output control and a commodity-row control can silently end up compared to each other.

Table 4: Structural differences between the reference Austrian calibration and the United States calibration used here. The right-hand column is not a list of approximations to be tightened later; several entries change what the model can be asked to do.

	Austria (reference)	**United States (this work)**
Input–output source	Eurostat 64-category NACE/CPA product-by-product tables at *basic prices*, already symmetric and balanced [Eurostat, 2025]	BEA 71-industry → 68-commodity make–use system, published at *purchasers' prices*, requiring a valuation bridge and an industry→commodity projection [U.S. Bureau of Economic Analysis, 2025a]
Firm counts	Business demography: active enterprises per sector	Census Statistics of U.S. Businesses [U.S. Census Bureau, 2025] plus USDA farm counts and pseudo-producers; a mixed enterprise/establishment ontology
Agent scale	Approximately 1:1 with observed entities	10^{-5}: 130 firms, $\approx 2\,831$ modelled people; **54 of 68 sectors hold exactly one firm**
Foreign block	Euro-area output and inflation are genuinely *exogenous* to Austria	The "euro-area" block is filled with *U.S.* data, so the Taylor rule reacts to an exogenous proxy for the same economy the model is simulating; there is no independent foreign demand state and no policy feedback loop
Labour and financial accounts	Harmonised Eurostat sector accounts, one vintage, one classification	CPS, CES and QCEW surveys stitched to the Financial Accounts (Z.1) across different statistical universes [U.S. Bureau of Labor Statistics, 2025, Board of Governors of the Federal Reserve System, 2025b]
Unemployment insurance	A single national replacement rate	State-level programmes varying by wage, duration and emergency extension; the Austrian constant $\theta^{UB} = 0.55(1-\tau^{INC})(1-\tau^{SIW})$ survives unchanged in the code
Trade closure	Imports read off the balanced table; the balance clears by construction	Imports measured from BEA T262 while the supply–use residual is *not* re-absorbed, so the residual survives as a signed sector-level gap
Structural vintage	Structure recalculated at or near each forecast origin	**One** 2024 structural row carried backwards to all 61 origins

One structural year for fifteen years of origins. The U.S. artifact contains 37 annual arrays of length one, all frozen at 2024: the entire input–output block, firm counts, employees, population, census unemployed and inactive, fixed assets, dwellings, capital consumption, every tax and benefit aggregate, and sectoral wages. Out of the box only nine origins are constructible. Unlocking the historical grid required a single-array rewrite — shallow-copying the calibration dictionary and setting the year index to the origin's year, which makes the calibration index resolve to 1, the only valid index for a length-one array. All quarterly series already run from 1996Q4, so opening *levels* do scale with the era (model opening real GDP is 5.483×10^6 at 2019Q4, 3.745×10^6 at 2010Q2 and 3.231×10^6 at 2005Q2). The consequence is a clean and declarable split, which the results tables respect:

- **Flow targets** — real GDP growth, deflator inflation, nominal GDP growth, the policy rate — are initialised from quarterly series with full history. The 2024 structure enters as a fixed technology and demand-composition matrix, not as a level. Scoring them is defensible as a diagnostic.
- **Stock-initialised targets** — unemployment above all — open at the 2024 level at *every* origin. Scoring them is not defensible, and §10.6 documents the failure quantitatively.

And therefore the trade-closure problem. Put the three previous paragraphs together. A published supply–use system balances by construction, but only in the valuation basis and on the axes in which it was published. Once the U.S. system has been bridged across valuation bases and projected from industries onto commodities, the balance is no longer guaranteed — and the artifact's decision to take imports from the measured T262 matrix, rather than solving them as the residual, removes the one remaining degree of freedom that could have absorbed the difference. The residual therefore survives into the simulator as a signed, sector-level gap. That is the defect, and §8 measures it.

5 Scientific assumptions register

Every assumption on which a reported number depends is listed here with its justification and its status. **[S]** marks a structural choice — one that restores an accounting identity or corrects an estimator that is wrong on its own terms, using past-only information. **[T]** would mark behavioural tuning against forecast performance; no entry carries it, and none was done.

A1. Revised-data protocol; current-vintage truth, symmetric across models. The scoring panel is a current-vintage snapshot of 101 quarters, 2000Q3–2025Q3, used for both estimation and truth by all fourteen models. Real GDP growth for 2010Q2 is the value as revised through 2026, not the value available in 2010. *Status:* a known limitation, symmetric across every model, and the same design the reference paper adopts for the bulk of its evaluation. *Flags:* `real_time = false`, `origin_admissible = false`, `promotion_eligible = false`.

A2. 2024 structure at every historical origin. The 37 frozen annual arrays constitute future information at every origin before 2024. *Status:* asymmetric and ours alone — the sharpest departure from the reference paper. Bounded for flow targets; fatal for stock-initialised targets, which is why unemployment is excluded from every weighted score.

A3. Agent scale 10^{-5}. 130 firms. *Justification:* an agent-count sweep from 130 to 2 708 firms leaves the truncated-output share at 14 % and mean growth near zero, so the central result is scale-invariant. *Status:* adequate for the mean, *inadequate for dispersion*: 54 of 68 sectors hold exactly one firm at this scale.

A4. Normal capacity utilisation $\omega = 0.85$ [S, inherited]. Hard-coded upstream. It fixes opening headroom at $1/0.85 - 1 = 17.6\,\%$ for every firm by construction. *Status:* an inherited magic number. Sourcing it from the Federal Reserve G.17 capacity utilisation series [Board of Governors of the Federal Reserve System, 2025a] would be a structural improvement; sweeping it to improve RMSE would be tuning and is prohibited.

A5. Opening inventories $S_i(0) = 0$ after reconciliation [S]. The first-pass artifact set $S_s = \tau \cdot \max(0, -\text{discrepancy})$, so only the 42 negative-gap sectors received an opening buffer and the 26 positive-gap sectors received exactly zero; the total was 7.4 % of quarterly gross output with 31 % of it in a single sector. *Justification:* a signed accounting residual is not an inventory stock, and the rule was asymmetric. After the balance clears there is no signed discrepancy to promote, and the reference specification opens with $S_i(0) = 0$.

A6. Production-account anchoring, $\lambda = 0.983460$ [S]. Biproportional reconciliation requires that row controls and column controls sum to the same total. In the artifact they do not: uses exceed supply by 1.0104 % in aggregate. That residual decomposes *exactly* — to 5.6×10^{-9} — into two measurement facts of the source data, shown in Table 5. One control must therefore move, and the repair anchors on the production account, scaling the four final-demand aggregates (and the two components that set the investment budget) by

$$\lambda \;=\; \frac{\text{supply} - \text{intermediates}}{C + G + I + X} \;=\; 0.983460.$$

Status: λ is fixed by the accounting identity alone and was not chosen with reference to any forecast error. The semantics are recorded verbatim in every result manifest. The alternative closure — hold every measured budget and let a uniform 1.01 % excess-demand wedge survive — was built and measured too; it overshoots, because a uniform 1 % excess demand is itself a growth impulse (§12.1).

A7. Random-walk-with-drift growth expectations [S]. The first-pass expectation is an OLS AR(1) with constant fitted to the *log level* of an I(1)-with-drift series. For a log-linear trend $x_t = a + gt$, OLS gives $\hat{\beta} = \bar{x}_t - \hat{\alpha}\bar{x}_{t-1}$ and hence the closed form

$$\gamma_e \;\approx\; g\Big[\,1 - (1 - \hat{\alpha})\tfrac{T}{2}\,\Big]. \tag{2}$$

The shrinkage is unavoidable: $\hat{\alpha}$ carries the Kendall small-sample downward bias [Kendall, 1954], and the level shocks of 2008–09 and 2020 push it further down. Shorter windows do not help — they trade a smaller T against a more biased $\hat{\alpha}$. Table 6 shows the estimator delivering 32–63 % of the in-sample trend at every origin, and Eq. (2) reproducing it to 0.05 pp. *Status:* a specification correction, estimated on past data only. Gated behind a model property that is *default-off*, so the Austrian and Italian calibrations remain bit-identical.

A8. Replacement-only investment [inherited, unrepaired]. $K' = K$ exactly when firms obtain desired investment and produce at target. *Status:* an upstream design property, not a fork bug, and *not* repaired here. It is the binding limitation on forward use (§12.1).

A9. No behavioural fitting. Verified numerically (§3); the forecast-error override layer is never read on the paths that produce these results.

A10. Unemployment excluded from every weighted score. Recorded in each manifest with the reason: the initial unemployed stock is a length-one annual array frozen at 2024, so every historical origin opens at the 2024 labour market. It is emitted and diagnosed (§10.6) but never scored.

A11. Seeds and Monte-Carlo policy. Simulation seeds are

$$\text{seed}(\text{origin}, \text{path}) \;=\; \text{hash}\big((\texttt{:abm\_revised\_v1}, \text{origin}, \text{path})\big) \bmod 2^{30},$$

so no two cells share a stream. The repaired model draws the *identical* seed stream as the first pass, which makes the contrast matched-seed; the expectations patch consumes exactly one Normal variate on either branch precisely to preserve that alignment. Failed paths are counted and left visible, never resampled; the count is zero everywhere.

At 2024Q4, Eq. (2) gives $(1 - 0.99305) \cdot 112/2 = 0.389$, hence $\gamma_e \approx 0.611\,g = 1.47\,\%$ against a measured 1.52 %.

6 Forecast design

6.1 Origins, horizons and the free-running protocol

Figure 1 shows the whole apparatus. Table 7 gives its settings.

Table 5: Exact decomposition of the aggregate excess of uses over supply in the first-pass artifact. Both terms are measurement facts of the source data, not model choices; neither can be removed without moving a measured control. Units are millions of current U.S. dollars.

Term	Value (US$m)	Meaning
Expenditure GDP − production GDP	442 933	the artifact's own statistical discrepancy, 1.57 % of GDP
Capital consumption − non-dwelling GFCF	88 223	net non-dwelling fixed investment is negative in the artifact
Total	**531 156**	= sum(uses) − sum(supply), matched to 5.6×10^{-9}

Table 6: The growth-expectation estimator, measured at six origins. T' is the length of the model's own output history at the origin; $\hat{\alpha}$ is the fitted AR(1) coefficient on the log level. The last column is the fraction of the in-sample trend that the expectation delivers. Pinning $\hat{\alpha} = 1$ — a random walk with drift, using the same past-only data — recovers +2.33, +2.39, +2.36, +2.42, +2.37 and +2.40 %: the trend, at every origin.

Origin	T'	$\hat{\alpha}$	γ_e (%)	trend (%)	γ_e/trend
2012Q4	64	0.9713	+0.745	+2.331	0.32
2014Q4	72	0.9791	+0.914	+2.385	0.38
2016Q4	80	0.9834	+1.035	+2.359	0.44
2019Q4	92	0.9906	+1.469	+2.415	0.61
2022Q4	104	0.9900	+1.240	+2.366	0.52
2024Q4	112	0.9930	+1.523	+2.401	0.63

6.2 Operators

Forecast targets are constructed *pathwise, before any ensemble operation.* Writing rg and ng for a path's real and nominal GDP series and taking row 1 as the origin, so that horizon h is row $h + 1$:

$$\text{real\_gdp} = 400\,\big(\log rg_{h+1} - \log rg_h\big), \tag{3}$$

$$\text{gdp\_deflator} = 400\,\big[(\log ng_{h+1} - \log rg_{h+1}) - (\log ng_h - \log rg_h)\big], \tag{4}$$

$$\text{nominal\_gdp} = 400\,\big(\log ng_{h+1} - \log ng_h\big), \tag{5}$$

$$\text{EFFR} = 100\,\big((1 + r_{h+1})^4 - 1\big). \tag{6}$$

The *GDP deflator* is the ratio of nominal to real GDP — the economy-wide price index implied by the accounts — so its log difference is nominal growth minus real growth. Writing it in log form rather than as a ratio is algebraically identical and finite for all positive floating-point inputs. The quarterly-to-annual-rate scaling cancels from log differences, so $400 = 4 \times 100$ is the complete annualisation. Unemployment has no field in the data structure and is read off agent state inside the stepping loop, with $H^{\text{act}} = 1\,826$ — the concept closest to the labour-force denominator — in the denominator.

6.3 Ensemble functionals

Two point forecasts are emitted per cell, as separate model columns: the *mean* of the pathwise-transformed values and the *median.* This follows Gneiting [2011]: the optimal point forecast depends on the loss function, the mean being the Bayes act under squared error and the median under absolute error. Reporting both is what allowed the first pass's mean-absolute-error weakness to be attributed correctly (§10) instead of being blamed on the choice of functional.

Table 7: Experimental design. "Free-running" means the model is initialised once at the origin and stepped forward twelve quarters with no re-anchoring, no observed-data injection and no conditioning on realised post-origin values. This corresponds to the reference paper's unconditional forecast, the harder of its two exercises.

Origins	61 (2010Q2–2025Q2)	Paths per origin	500
Minimum training window	40 quarters	Path failures	0
Horizons scored	1, 2, 4, 8, 12	Simulation mode	free-running, serial
Simulated quarters per path	12	Model scale	10^{-5}
Burn-in quarters	0	Julia / threads / BLAS	1.10.3 / 1 / 1

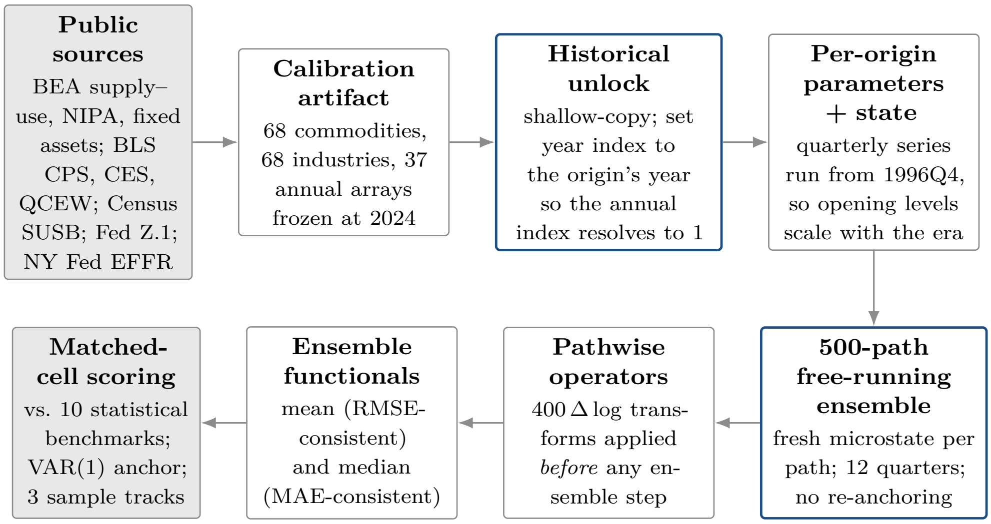

Figure 1: F7 — the forecasting pipeline. Public source data are compiled once into a calibration artifact whose annual structural block is a single 2024 row. The historical unlock re-points the annual index so that the artifact can be instantiated at any origin; the quarterly series, which run from 1996Q4, then supply era-appropriate opening levels. Each origin is simulated 500 times, transformed pathwise, reduced to a mean and a median point forecast, and scored on cells shared with every statistical benchmark.

6.4 The comparator field

Fourteen model columns are scored: ten statistical benchmarks plus mean and median columns for each of the two ABM versions. Table 8 gives the specifications, taken from the benchmark module's own type definitions.

The VAR(1) is the ratio anchor: every reported ratio is a model's score divided by the VAR(1)'s score on the same cell, so values below 1.0 mean the model beats the anchor. The anchor is not a strong benchmark — at the 2010Q2 origin its estimated companion matrix has spectral radius 1.0342 and design condition number 8 747, i.e. it is borderline explosive and poorly conditioned — and this should temper any reading of the absolute level of a ratio, though not of the differences between models scored against the same anchor.

A fixed-parameter semi-structural state-space comparator (potential growth, output gap, natural unemployment, neutral real rate, inflation anchor, inflation, policy rate and lagged output gap, with a fixed transition encoding IS, Phillips, Okun and Taylor links, updated by an exact Kalman filter) exists in the repository but is *not* in this field: it scores a core-four target set of which the ABM cannot serve the personal consumption expenditures price index. Its standing from a separate 11-model comparison is a weighted RMSE ratio of 0.749 (1 of 11, all-available) and 0.745 (1 of 11, balanced), quoted as context only, on a different target set and a different field.

Table 8: The ten statistical benchmarks. All are estimated on training data only at each origin, with hyperparameters fixed before the fit and encoded in the model identifier; none is tuned inside the evaluation.

#	model_id	Specification
1	`naive_no_change`	Random walk: forecast equals the last training observation.
2	`naive_drift`	Random walk with drift estimated from training differences only.
3	`naive_historical_mean`	Constant forecast equal to the training-sample mean.
4	`univariate_ar_p1_constant`	Univariate AR(1) with intercept, per target.
5	`univariate_ar_p4_constant`	Univariate AR(4) with intercept.
6	`univariate_ar_bic_...`	Lag order chosen by the Bayesian information criterion on a common, training-only window.
7	`beforeit_var_p1_constant`	Multivariate VAR(1) with intercept over the eight-target panel. **Ratio anchor.**
8	`beforeit_var_p2_constant`	VAR(2).
9	`beforeit_var_p3_constant`	VAR(3).
10	`bvar_mniw_v1_p1_...`	Natural-conjugate Bayesian VAR with a matrix-normal / inverse-Wishart prior and Minnesota-style shrinkage [Litterman, 1986]: tightness 0.2, lag decay 1.0, own-lag prior mean 0.0, intercept variance 100, inverse-Wishart degrees-of-freedom offset 2, innovation scale 1.0, scale floor 10^{-8}.

6.5 Truth, scaling and score construction

Truth is the revised panel itself; the error sign convention is forecast minus actual. Target transformations are *not* uniform across the panel: the four GDP-family series are $400 \ln(\text{level}_t/\text{level}_{t-1})$; the unemployment rate is a quarterly mean of three monthly levels with no transform; EFFR is the equal-weight mean of published business-day rates within the quarter [Federal Reserve Bank of New York, 2025].

Three scores are reported. *RMSE* (root mean squared error) penalises large errors quadratically; *MAE* (mean absolute error) penalises them linearly; *MASE* (mean absolute scaled error, Hyndman and Koehler, 2006) divides MAE by the in-sample mean absolute first difference of the same series, making it scale-free and directly comparable across targets. MASE scales are recomputed at every origin from training data only.

The *weighted ratio* used for the standings is a macro-average of matched cellwise ratios,

$$\mathcal{W} = \sum_{\text{targets}} \sum_{h} w_{\text{target}}\, w_h \,\frac{\text{score}_{\text{model}}(\text{target}, h)}{\text{score}_{\text{anchor}}(\text{target}, h)}, \qquad w_h \in \{0.30, 0.25, 0.20, 0.15, 0.10\}, \quad (7)$$

with equal target weights and horizon weights declining in h. It is explicitly *not* a ratio of pooled losses, which would let one badly scaled target dominate. A model is marked `INCOMPLETE_MATCHED_GRID_NOT_RANKED` unless it has every target-by-horizon cell and zero failures; every ABM column here is `COMPLETE_MATCHED` at 10 of 10 cells on all tracks and both target sets.

The scoring key is (origin, target, horizon), and only cells where *every* model in the field is present survive. Because both ABM versions were scored in a single run, the version contrast cannot be a sample artifact.

Table 9: The three sample tracks and their common-cell counts. The pandemic mask is applied to *target* periods, not origins, and was pre-registered in the run manifest rather than selected after seeing results.

Track	Rule	n at $h = 1/2/4/8/12$
all-available	every common cell	61 / 60 / 58 / 54 / 50
balanced h=12	origins with a complete 12-quarter horizon	50 / 50 / 50 / 50 / 50
pandemic-masked	exclude cells whose target period falls in 2020Q1–2021Q4	53 / 52 / 50 / 46 / 42

6.6 Sample tracks and the pandemic mask

6.7 Monte-Carlo error

The ABM's point forecast is an ensemble functional over finitely many stochastic paths, so its measured RMSE is inflated by approximately $\sqrt{1 + (\mathrm{se}_{\mathrm{MC}}/\mathrm{RMSE})^2}$. Table 10 shows that at 500 paths this is negligible: against a real-GDP cross-origin RMSE of order 6 pp, a Monte-Carlo standard error of 0.14 pp inflates the measured RMSE by well under 0.1 %. (At the 16-path pilot count the standard error was 1.3 pp and *was* material; 500 paths were used precisely to remove the confound.)

Table 10: Monte-Carlo error at 500 paths, ranges over the five horizons. "Ensemble s.d." is the dispersion of the simulated paths — genuine model uncertainty; "MC s.e." is the sampling error of the ensemble mean — simulation noise.

Target	Mean ensemble s.d.	Mean MC s.e. of the mean	Max MC s.e.
`real_gdp`	3.036–3.269	0.136–0.146	0.255
`gdp_deflator`	0.860–1.077	0.038–0.048	0.100
`nominal_gdp`	3.211–3.440	0.144–0.154	0.264
`unemployment_rate`	0.414–1.079	0.019–0.048	0.070
`effective_federal_funds_rate`	0.139–0.336	0.006–0.015	0.019

6.8 The opening-row measurement basis

One artifact of initialisation must be declared because it affects the $h = 1$ cell, which carries the largest horizon weight. The initialiser writes row 1 on a different measurement basis from rows 2 onwards: prices are normalised to 1 at initialisation, so real and nominal GDP coincide in row 1 and the model-implied opening deflator is exactly 1.0. The opening row is the *model-implied* opening, not an observed control — reassuring for leakage, unhelpful for $h = 1$ comparability. Dropping $h = 1$ was rejected outright: it would render the grid incomplete and the model unranked. It is reported and flagged.

7 Results I: the first pass

All results in §7–§10 are computed from the joint scoring run in which both ABM columns were scored together on identical cells. The first-pass column reproduced the earlier standalone run bit-identically over all 3 660 cached cells (gate G4 in Table 18).

7.1 Standings

Competitive on the first two tracks — and eleventh of fourteen once the COVID cells are removed. That regime dependence was, at the time, the most alarming feature of the result: a

Table 11: First-pass (`v1`) standings on the headline target pair {real GDP growth, GDP-deflator inflation}. Ranks are out of the 14 scored forecast columns.

Track	v1 mean ratio (rank)	v1 median ratio (rank)	Best statistical
all-available	0.905 (5th)	0.903 (4th)	`naive_historical_mean` 0.879
balanced h=12	0.894 (5th)	0.893 (4th)	`naive_historical_mean` 0.884
pandemic-masked	1.272 (11th)	1.264 (10th)	`univariate_ar_p1` 0.848

model whose standing depends on the presence of the two most anomalous years in the sample is not obviously forecasting anything.

7.2 The bias

Table 20 (§10, left-hand columns) gives the detail. Real GDP growth is under-forecast by -1.32, -7.28, -2.94, -2.15 and -2.02 pp at $h = 1, 2, 4, 8, 12$. The -7.28 pp figure at $h = 2$ is a transient; the -2 to -3 pp figures at the longer horizons are a chronic level bias. The deflator, by contrast, was already the strongest headline target, with ratios of 0.771–0.975 at every horizon and MAE ratios uniformly below one.

7.3 The mean-absolute-error asymmetry

The first pass's headline RMSE ratio of 0.905 sat against an MAE ratio of 1.137 — it beat the anchor under quadratic loss and lost under linear loss. Disaggregating shows the asymmetry was never general: on the deflator and the policy rate the MAE ratios were already below one. It was concentrated in real and nominal GDP and, within those, in the $h = 2$ cell (MAE ratios 2.376 and 2.071 respectively). Switching to the loss-consistent median improved the weighted MAE ratio only from 1.137 to 1.130, so it was *not* a functional-choice artifact.

7.4 Burn-in: a negative result, retained

Burn-in — building the model k quarters before the origin and treating row $k + 1$ as the origin — was the leading candidate fix for the $h = 2$ transient. It was tried at $k = 1$ and $k = 4$ with 128 paths, and it failed in an instructive way. One quarter of burn-in *relocates* the transient from $h = 2$ to $h = 1$, where the horizon weight is larger, so the weighted result worsens (0.905 $\rightarrow$ 0.954). Four quarters disperses the transient (headline MAE ratio improves to 1.012) but destroys the level targets: the policy-rate $h = 1$ ratio degrades from 0.563 to 2.091 and the secondary weighted ratio from 0.782 to 1.095, because four quarters of free-running drift move the model's policy rate away from the origin's observed level before scoring begins. Burn-in trades a growth-rate artifact for a level-drift information loss.

This entire line of attack is moot after the repair — the $h = 2$ bias falls to -0.096 pp under the balance fix alone, with zero burn-in — and the burn-in runs are retained only as documented negative results.

8 Diagnosis: the anatomy of a -2.75 percentage-point bias

This section rests on matched-seed A/B experiments at four to five origins with 48–64 paths per cell, with seeds identical across every variant. The bias measured here is $h = 3 \ldots 12$ annualised real GDP growth, ensemble mean minus realised, averaged over three clean origins (2012Q4, 2016Q4, 2022Q4; 2019Q4 is excluded from every average because its window is the pandemic rebound at +6.2 pp). The baseline is -2.750 pp, consistent with the -2.0 to -2.9 pp per-horizon biases measured independently at 61 origins.

8.1 Ranked attribution

Table 12: Ranked attribution of the −2.75 pp real-growth bias.

#	Mechanism	Share	Confidence
1	Non-clearing sector commodity balance under explicit measured trade, colliding with a frozen firm capacity envelope	−2.10 pp (76 %)	high
2	Growth expectation from an OLS AR(1) on log levels, delivering 32–63 % of the in-sample trend	−0.70 pp (24 %)	high (estimator), medium (delivered size)
3	Opening inventories built from the negative half of a signed statistical discrepancy	−0.23 pp (a subset of #1)	high
4	No capital accumulation (replacement-only investment)	latent; binds only through #1	high
5	Exogenous government/export/import log-level AR(1) attractors	≈ 0 net	high

The GDP deflator is untouched throughout — 1.8–2.5 % in every variant — which is why inflation was never biased. **The defect is a real quantity-rationing defect, not a pricing defect.** §10 confirms this prediction out of sample.

8.2 The shortfall is broad-based

Decomposing the expenditure side (contribution gap = opening nominal share × (model − realised)) shows every domestic component growing at 0.1–0.5 % annualised against realised 2.5–5.4 %:

Table 13: Expenditure decomposition, $h = 3 \ldots 12$ annualised; contribution gaps in percentage points of GDP growth. Consumption carries ≈ 70 % of the gap and fixed investment ≈ 25 % *purely because of their weights* — their growth rates are equally flat (0.11–0.36 %). Net trade contributes positively; exports are over-forecast by 1.3–3.0 pp.

Origin	C	I fixed	G	X	M (−)	**GDP**
2012Q4	−1.77	−0.91	−0.02	+0.23	+0.44	**−2.03**
2014Q4	−1.54	−0.64	−0.17	+0.10	+0.46	**−1.80**
2016Q4	−1.54	−0.60	−0.46	+0.12	+0.19	**−2.30**
2022Q4	−1.72	−0.44	−0.30	−0.05	+0.30	**−2.20**

At the 68-sector level, 31–36 of 68 sectors show positive growth, and *the same sectors lose at every origin* — the persistent-loser list is identical at 2012Q4, 2016Q4 and 2024Q4. A stable loser set across twelve years of origins is a calibration signature, not a shock response.

8.3 Defect 1: the commodity balance does not clear

Because imports come from the measured BEA T262 matrix rather than being solved as the supply–use residual, the residual survives as a signed, sector-level gap. Measured on the shipped first-pass artifact:

$$\frac{\text{aggregate uses}}{\text{aggregate supply}} = 1.01010,$$

$$\min -0.706 \;\mid\; p_{25} - 0.058 \;\mid\; \text{med.} + 0.020 \;\mid\; p_{75} + 0.128 \;\mid\; \max +0.476,$$

with 42 of 68 commodities over-demanded, 34 with |gap| > 10 % and 8 with |gap| > 25 % (mean |gap| 0.1262, standard deviation 0.1790). The left panel of Figure 3 shows the distribution.

Two facts then lock the chain, and Figure 2 draws it.

1. **The capacity envelope is fixed by construction.** $\omega = 0.85$ is hard-coded, and the capital calibration gives $K_i\kappa_i = Y_i/0.85$ identically, so every firm opens at exactly 85 % utilisation with 17.6 % headroom, in proportion to the *opening output* composition.
2. **Capital never accumulates.** Replacement-only investment gives $K' = K$ exactly when firms obtain desired investment and produce at target. Measured $K_{\text{end}}/K_1 = 1.0000$–$1.0009$ in every variant, *including* under a 30 % capacity relief.

The demand vector, meanwhile, is fixed by the coefficient matrices, and the two differ by exactly the 42/26 signed gap. Over-demanded sectors sit permanently against their ceiling, under-demanded ones permanently idle, and the min in Eq. (1) is one-sided so the idle capacity never compensates. The signature matches the mechanism precisely:

- mean (uses – supply)/supply of the twelve most-truncated sectors is +0.222, against +0.014 for all 68;
- mean gap of the persistent value-added losers is −0.140, against −0.025 for the winners;
- 46 of 68 sectors are *never* truncated, while a stable minority is truncated in 40–80 % of quarters;
- 12.9–14.4 % of gross output is produced under a binding ceiling every quarter, at every origin tested (Table 16, last column).

Downstream, the unmet demand appears as persistent rationing, stable at every horizon:

Table 14: Fill rates — the fraction of each buyer's demand actually satisfied — in the first-pass model. Exports are the worst-served buyer because the export basket is concentrated in over-demanded commodities. Relieving opening capacity by 30 % lifts the export fill rate to 0.92–0.96, which establishes that export rationing is a *symptom* of the capacity/balance collision rather than an independent trade defect.

Fill rate	$h = 1$	$h = 2$	$h = 4$	$h = 8$	$h = 12$
Households	0.999	0.973	0.978	0.976	0.975
Government	0.999	0.961	0.961	0.960	0.958
Exports	0.995	**0.845**	**0.869**	**0.865**	**0.862**
Imports	0.993	0.991	0.989	0.989	0.987

8.4 Defect 2: the growth expectation is biased low by construction

The second defect is the estimator of Eq. (2) and Table 6. What makes it interesting is that it is *latent*. Replacing the model's opening output history by its own log-linear trend — a past-only intervention — raises γ_e from ≈ 1.0 % to ≈ 1.9 % but moves delivered GDP growth by +0.06 pp on average. Pushing the expectation to twice trend still delivers between −0.06 and +0.04 pp. The elasticity of delivered growth to γ_e is ≈ 0 while the imbalance is present, rises to $\approx +0.5$ pp once capacity headroom is opened, and reaches +0.75 pp once the balance is closed. Measured on the actual patched code with 64 matched-seed paths at four origins, the random-walk-drift change is worth +0.193 pp on the unreconciled baseline and +0.499 pp on the reconciled one — the monotone pattern the diagnosis predicted. **Expectations are a real but strictly subordinate defect: fixing them is worth +0.75 pp, but only after fixing the balance.**

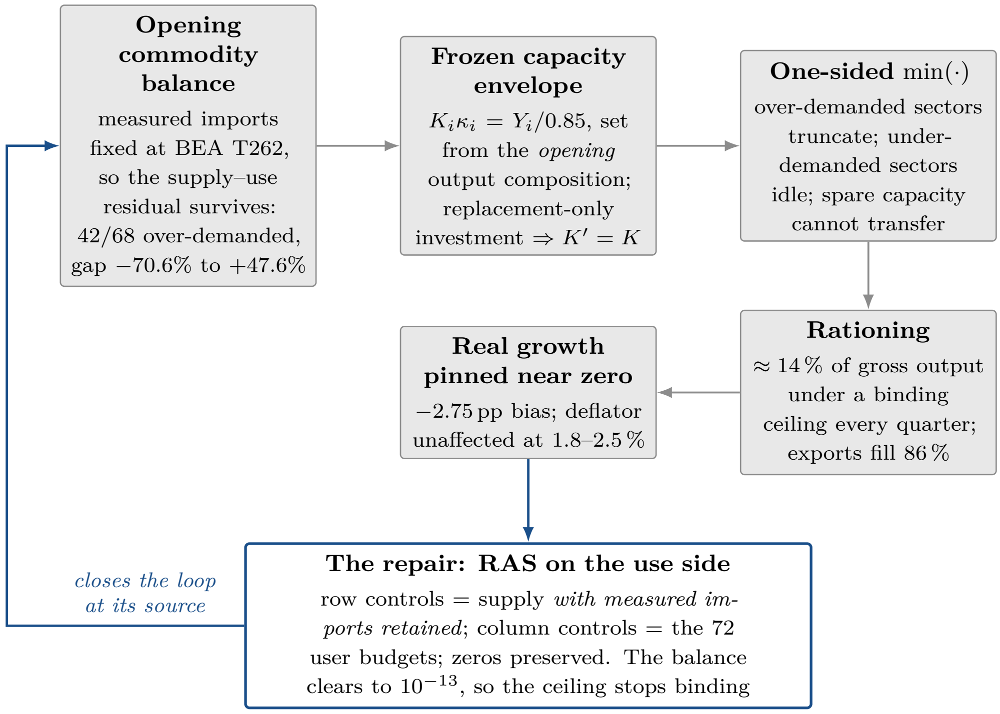

Figure 2: F8 — the defect mechanism and its repair. Grey: the causal chain from a non-clearing opening balance to a real-growth bias. Blue: the reconciliation closes the loop at its source, by making the opening balance clear commodity by commodity while retaining every measured import level. Because capacity relief and balance closure were shown to be *substitutes* rather than complements (Table 15), intervening at the balance is sufficient; the frozen capacity envelope is left untouched.

8.5 The falsification table

Each row of Table 15 is a matched-seed experiment run to kill a competing hypothesis. The sixth row is the decisive discriminator: capacity relief and balance closure are *substitutes*, not complements, which places the balance upstream.

8.6 The dose–response that motivated the repair

Reverting to the legacy residual closure — solving imports as the supply–use residual, which *guarantees* clearing but discards the measured BEA import levels — gives, for $h = 3 \ldots 12$ annualised real GDP growth:

Table 15: Falsification experiments. All matched-seed, all at scale 10^{-5} unless stated.

Test	Result	Rules out
Agent-count sweep, 130 → 294 → 912 → 2 708 firms	truncated share 14 % at *every* scale; growth stays ≈ 0	idiosyncratic small-sample lumpiness
Materials-only relief, $M \times 1.30$	−0.06 against −0.08 baseline: no effect	materials as an independent binding constraint
Investment-demand cut, $\kappa_s \times 1.30$ (cuts I^d by 23 %, leaves $K\kappa$)	−0.20 against −0.08: −0.12 pp only	investment demand as a first-order channel
Capacity tightening, $K\kappa \times 0.77$	−1.14 / −1.22 pp	— confirms the sign and convexity of the capacity margin
Un-capped investment emulation	−0.03 against −0.08; $K_{\text{end}}/K_1 = 1.0005$	"the fork's min broke investment"
Legacy residual closure + $K \times 1.30$	**1.921 vs 1.879 (2016Q4); 2.505 vs 2.503 (2024Q4): +0.00 pp**	**capacity as an *independent* cause**
Drop opening inventories only	+0.23 pp mean, same sign at all four origins	— a genuine but small independent contribution
Labour-supply ceiling	labour binds in 0.000 of rows at every origin	the labour block as the constraint
Post-origin look-ahead in the exogenous arrays	bit-identical paths under -10^{99} poisoning	data leakage

Table 16: Dose–response. The legacy residual closure is a *diagnostic instrument*, not the shipped repair: it discards measured import levels. The shipped repair achieves exact clearing while retaining them.

Origin	Realised	Baseline	Legacy closure	+ trend exp.	Truncated share
2012Q4	**2.586**	0.051	**2.365**	3.511	0.144 → 0.006
2016Q4	**2.922**	−0.079	**1.879**	2.633	0.139 → 0.002
2022Q4	**2.531**	−0.207	**1.813**	2.458	0.129 → 0.004
2024Q4	—	−0.219	**2.503**	2.623	0.143 → 0.012
Mean bias over the three clean origins: −2.76 pp → −0.66 pp → +0.19 pp, deflator unchanged throughout.					

9 The repair

Two changes, both in the calibration artifact rather than in behavioural parameters.

9.1 (i) The opening commodity balance now clears

A *biproportional* adjustment — the RAS method of Stone [1961] and Bacharach [1970], equivalently the minimum-I-divergence projection — finds the matrix closest to a given one, in a relative-entropy sense, subject to prescribed row and column totals. It does so by alternately scaling rows and columns until both sets of margins are satisfied.

Formally, let $U^0 \in \mathbb{R}^{G\times J}_{\geq 0}$ be the observed use matrix over $G = 68$ commodities and $J = 72$ users (68 industry intermediate columns plus C, G, I and X). We seek

$$U^\star \;=\; \arg\min_{U\geq 0} \sum_{g,j} U_{gj} \log \frac{U_{gj}}{U^0_{gj}} \quad \text{s.t.} \quad \sum_j U_{gj} = r_g \;\; \forall g, \qquad \sum_g U_{gj} = c_j \;\; \forall j, \tag{8}$$

with row controls r_g = industry output$_g$ + measured imports$_g$ and column controls c_j the 68 industry intermediate budgets and the four final-demand budgets. The solution has the biproductive form $U^\star_{gj} = \rho_g\, U^0_{gj}\, s_j$ and is obtained by iterative proportional fitting; structural zeros are preserved exactly, because a zero cell stays zero under any scaling.

The concrete run is reported in Table 17. Crucially, `use_explicit_trade` stays `true`: no measured import level is discarded. The reconciliation works *around* the measured trade data rather than throwing it away, which is what distinguishes the shipped repair from the legacy-closure diagnostic instrument of Table 16.

Table 17: The RAS reconciliation as executed, from the committed reconciliation report. The adjustment moves composition, not aggregates: every column budget is preserved to 10^{-13} after the λ scaling.

Flow system	68 commodities × 72 users
Structural zeros preserved	1 549 / 4 896 cells (31.6 %)
Row controls	industry output + measured imports, sum 5.25699×10^7
Column controls	measured budgets, held exactly, sum 5.25699×10^7
Convergence	623 iterations; final max relative control error 9.996×10^{-14}
Balance after RAS	$\max \lvert \text{uses}_g/\text{supply}_g - 1 \rvert = 5.55 \times 10^{-16}$; cross-sector s.d. 1.96×10^{-16} (was 0.1790)
Balance after re-derivation	$\max \lvert \text{uses}_g/\text{supply}_g - 1 \rvert = 1.00 \times 10^{-13}$
Total use adjustment	$\sum \lvert d \rvert = 6.786 \times 10^6$ (12.78 % of total uses)
Cellwise adjustment	$\sum \lvert dU \rvert = 7.423 \times 10^6$ (13.98 % of total flow)
Coefficient movement received by the simulator	
Technology matrix a_{sg}	$\max \lvert \Delta a \rvert = 0.1408$; mean $\lvert \Delta a \rvert = 0.00226$
Household basket b_g^{HH}	$\max \lvert \Delta \text{share} \rvert = 0.0158$; L_1 distance 0.0970
Government basket c_g^G	$\max \lvert \Delta \text{share} \rvert = 0.1450$; L_1 distance 0.3204
Investment basket b_g^{CF}	$\max \lvert \Delta \text{share} \rvert = 0.0401$; L_1 distance 0.1049
Export basket c_g^E	$\max \lvert \Delta \text{share} \rvert = 0.0309$; L_1 distance 0.1197
Import basket c_g^I	unchanged

The adjustment is not uniform across users: as a share of its own budget, the government column moves most (31.0 %), then intermediates (15.3 %), exports (12.1 %), investment (10.0 %) and households (9.8 %). The largest single commodity adjustments are professional and technical services (uses cut 32.3 %, closing a +47.6 % gap), wholesale trade (raised 42.5 %, closing a −29.8 % gap) and state and local government (raised 26.2 %). Figure 3 shows the before-and-after distribution.

The one accounting choice — production-account anchoring by $\lambda = 0.983460$ — is stated in assumption A6 and Table 5, and its semantics are sealed verbatim in every result manifest.

9.2 (ii) Growth expectations become a random walk with drift

The repaired model pins $\hat{\alpha} = 1$ and takes the drift as average past log growth — the correctly specified estimator for an I(1)-with-drift series, on past data only. It is gated behind a new model property, registered only when the calibration artifact carries the corresponding marker, and *default-off*, so the Austrian and Italian calibrations remain bit-identical (gate G3). Exactly one Normal variate is consumed on either branch, which keeps the matched-seed comparison aligned.

9.3 Regression gates

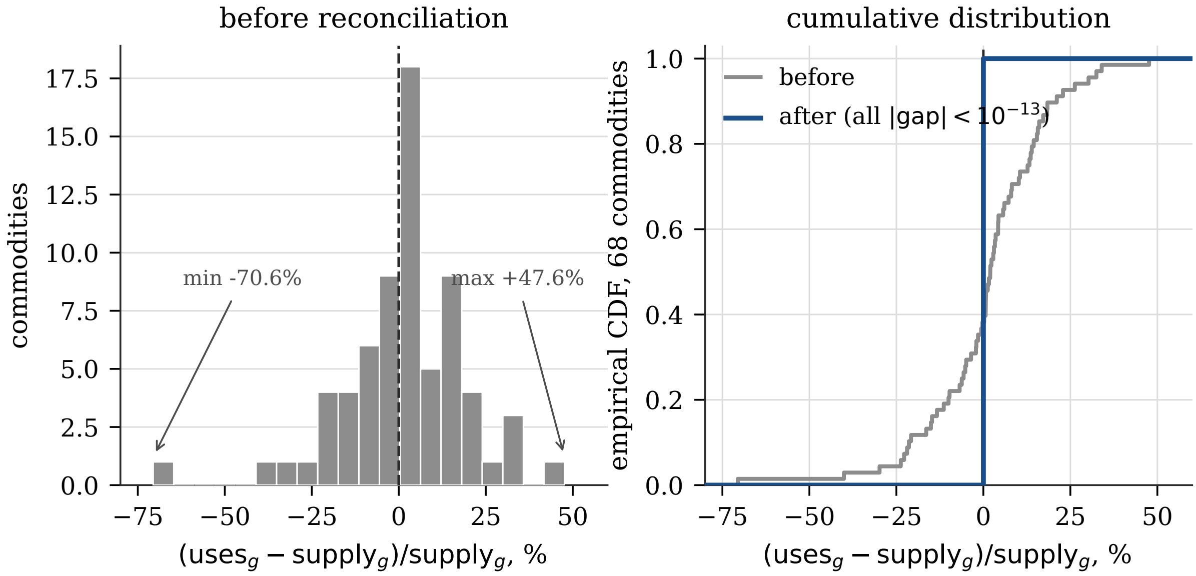

Figure 3: F5 — the opening commodity balance, before and after reconciliation. Left: the distribution of $(\text{uses}_g - \text{supply}_g)/\text{supply}_g$ over the 68 commodities of the first-pass artifact, spanning $-70.6\,\%$ to $+47.6\,\%$ with a standard deviation of 17.9 percentage points. Right: the empirical cumulative distribution before (grey) and after (blue). The post-reconciliation distribution is a step function at zero: every commodity clears to within 10^{-13}. Data: `reconciliation_by_commodity_rho1.csv`.

Table 18: Regression gates on the repair. G7 is a pre-existing repository-wide formatter failure on files untouched by this work, tracked separately. G8 is reported as failing on the merits rather than hidden.

#	Gate	Result
G1	matched-grid completeness	**PASS** — 61/61 origins; counts [61, 60, 58, 54, 50], [50, 50, 50, 50, 50], [53, 52, 50, 46, 42]; every ABM column `COMPLETE_MATCHED`
G2	no non-finite output	**PASS** — 0 non-finite cells in $3\,660 + 3\,660 + 120$ ensemble rows; 500 paths used at every origin; 0 failed; the failures file is empty
G3	Austria/Italy unchanged with the flag off	**PASS** — 3 seeds × 3 series, SHA-256 of raw `Float64` vectors identical between a pristine checkout and the patched worktree
G4	first-pass column reproduces its own earlier run	**PASS** — all 3 660 cached cells identical ($\max|\text{diff}| = 0$); weighted scores match to every printed digit
G5	GDP-deflator inflation stays $\approx 2\,\%$	**PASS** — 2.019 pp (p5 1.663, p95 2.687) against 2.024 pp over the same 671 scored cells; *bit-identical at* $h = 1$
G6	commodity balance clears in the shipped artifact	**PASS** — 1.0×10^{-13} after re-derivation through the unmodified pipeline
G7	package test suite	**PARTIAL** — 815 pass, 1 fail; the failure is a repository-wide formatter check that already failed at the base commit on ten files untouched here
G8	unemployment	**REPORTED, FAILING ON MERIT** — §10.6

10 Results II: the repaired model

10.1 Standings: first of fourteen in all three tracks

On the *balanced* h=12 track ($n = 50$ everywhere) the repaired mean column scores 0.829 (1st), its median 0.830 (2nd), `naive_historical_mean` 0.884 (3rd), the first-pass columns 0.893 and 0.894 (4th, 5th), and the BVAR 0.909 (6th).

On the *pandemic-masked* track ($n = 53/52/50/46/42$) the repaired mean column scores 0.796 (1st) and its median 0.799 (2nd), ahead of `univariate_ar_p1` at 0.848, the BVAR at 0.852, `univariate_ar_bic` at 0.870, `naive_historical_mean` at 0.881 and `univariate_ar_p4` at 0.889; the first-pass columns fall to 1.264 and 1.272 (10th, 11th).

The regime-sensitivity finding resolves in the repaired model's favour. The first pass's competitive rank *depended* on the COVID cells: masking 2020Q1–2021Q4 moved it from 5th (0.905) to 11th (1.272). The repaired model carries no such dependence — it is 1st in the masked sample and by its *widest* margin (0.796 against 0.848 for the best statistical model, a 6.1 % gap, against 5.6 % all-available). The regime sensitivity was a symptom of the defect: in the calm sample the chronic −2 pp bias dominated the error, while in the full sample it was partly hidden by cells that no model forecasts. Removing the bias removes the sensitivity.

10.2 Per-cell detail

Four readings of Table 20 and Figures 5–6.

1. **The real-growth bias is essentially eliminated.** $-1.32/-7.28/-2.94/-2.15/-2.02$ pp becomes $-0.03/-0.10/-0.14/-0.12/-0.20$ pp. The residual is under a quarter of a percentage point at every horizon, and its mild negative drift at $h = 12$ is consistent with the headroom mechanism of §12.1. In scale-free terms, real-GDP MASE moves from 1.259/3.361/1.819/1.629/1.656 to 1.043/1.099/1.137/1.221/1.259.
2. **The $h = 2$ transient is gone**, with zero burn-in. This retires the burn-in hypothesis of §7: the opening-row measurement basis is real, but it accounted for a small share of the −7.3 pp. The bulk was the rationing crash, and it disappears when the balance clears.
3. **The deflator is untouched** — bit-identical at $h = 1$, within 0.006 of the first-pass ratio at $h = 12$, and slightly *worse* in rank at $h = 2$ and $h = 4$ only because the repair moved the models around it. This is the diagnosis's central prediction confirmed out of sample: a quantity defect, not a pricing defect.
4. **The policy rate is essentially unchanged**, with ratios differing in the fourth decimal. The Taylor rule is fed a corrected output path, but the rate was never the problem.

10.3 The mean-absolute-error asymmetry is resolved, not traded away

The weighted MAE ratio moves from 1.137 to **0.814**. Per cell:

10.4 Density calibration

A point forecast is only part of what a simulation ensemble produces. *Interval coverage* asks how often the realised value falls inside a stated predictive interval: a well-calibrated 90 % interval should contain the truth 90 % of the time. Table 22 and Figure 7 report coverage pooled over all five horizons ($n = 283$ per target).

The repaired model's real-GDP density is well calibrated at the 90 % and 80 % levels (0.929 and 0.894 against nominal 0.90 and 0.80). The first pass's 0.654 and 0.580 were *not* a width

Table 19: Headline standings, {real GDP growth, GDP-deflator inflation}, *all-available* track ($n = 61/60/58/54/50$). Ratios are weighted macro-averages of matched cellwise ratios against the VAR(1) anchor; lower is better. First-pass rows are italicised.

Rank	Model	Weighted RMSE ratio	Weighted MAE ratio
1	**ABM v2, mean**	**0.830**	**0.813**
2	**ABM v2, median**	**0.830**	**0.815**
3	`naive_historical_mean`	0.879	0.838
4	*ABM v1, median*	*0.903*	*1.130*
5	*ABM v1, mean*	*0.905*	*1.137*
6	`bvar_mniw_v1_p1_constant`	0.908	0.896
7	`univariate_ar_p1_constant`	0.918	0.888
8	`univariate_ar_bic_...`	0.929	0.901
9	`univariate_ar_p4_constant`	0.939	0.923
10	`beforeit_var_p1_constant` (anchor)	1.000	1.000
11	`naive_no_change`	1.075	1.121
12	`naive_drift`	1.106	1.156
13	`beforeit_var_p2_constant`	2.121	1.496
14	`beforeit_var_p3_constant`	5.313	2.672

problem — the intervals missed because the point forecast was biased down by ≈ 2 pp. Nominal GDP improves comparably. The 50 % band now over-covers (0.696 against 0.500): the centre of the repaired ensemble is too wide.

Three targets remain badly calibrated and are named as open work. The deflator is under-dispersed at every level and *unchanged* by this work — the ensemble's inflation spread is genuinely too narrow. The policy rate under-covers badly and increasingly with horizon (0.754 at $h = 1$ down to 0.200 at $h = 12$) in both versions — the Taylor-rule path is far too tight. Unemployment coverage *falls* and reaches exactly 0.000 at $h = 12$, which is the defect of §10.6 appearing in the density as well as the mean. This is a coverage assessment only: no probability-integral-transform histogram, log score or continuous ranked probability score was computed, so it is not a full density evaluation.

10.5 Secondary targets

Nominal GDP growth is the exact sum of real GDP growth and deflator inflation, so it adds no independent information but costs nothing to report. The policy-rate operator maps the model's internal Taylor-rule rate to an annualised percentage point; **this is explicitly not an approved EFFR bridge**, and the caveat travels with every number in Table 23.

The pandemic-masked loss is reported as a loss. On the secondary pair in the calm sample, AR(4) at 0.719 beats the repaired ABM at 0.739 — a 2.8 % margin. The repaired model wins the other two tracks and the headline pair everywhere, but it does not sweep the field, and the one cell where a statistical model wins outright is stated here rather than buried.

10.6 Unemployment: an open defect that became worse

`unemployment_rate` is excluded from every weighted score in both versions, for the reason recorded in each manifest: the initial unemployed stock is a length-one annual array frozen at 2024, so every historical origin opens at the 2024 labour market.

The first-pass defect was a constant forecast. Across all 61 origins the $h = 1$ unemployment forecast ranges from 3.492 % to 3.679 % — a span of 0.187 pp, cross-origin standard deviation 0.048 pp — against realised values spanning 3.533 % to 13.000 %. Had it been scored, the first pass would have ranked 1st or 2nd of the 14 columns at $h = 8$ and $h = 12$. *Those*

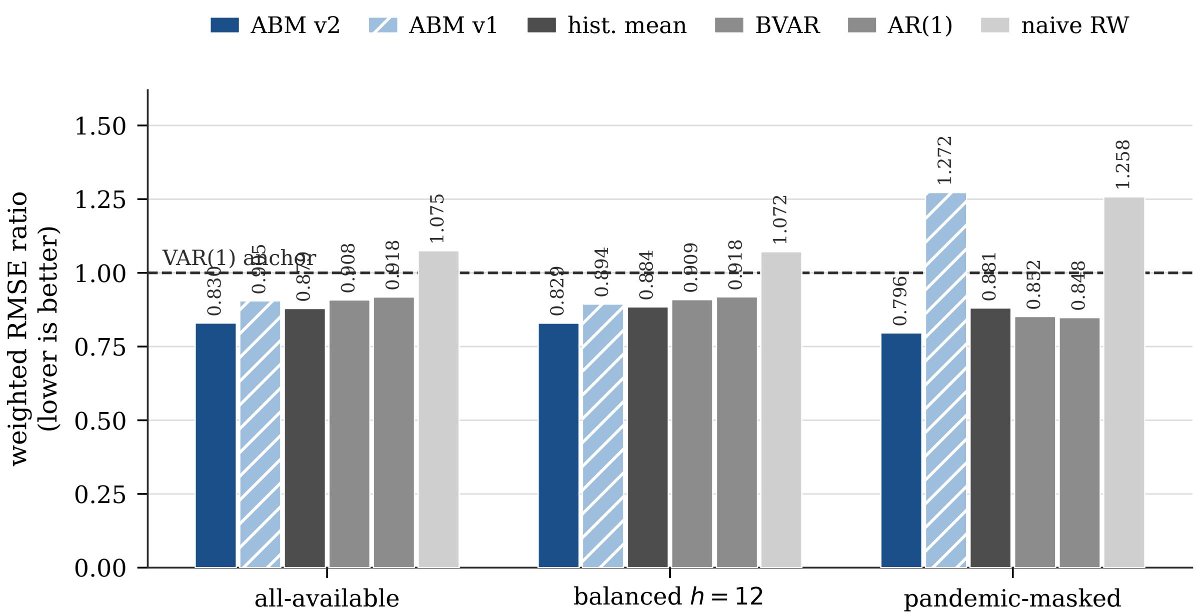

Figure 4: F1 — weighted RMSE ratio on the headline target pair, three sample tracks. Blue is the repaired ABM; hatched pale blue is the first pass; greys are statistical benchmarks. The dashed line is the VAR(1) anchor at 1.0. The repaired model is first in all three tracks, and its margin is *widest* in the pandemic-masked sample — the opposite of the first pass's behaviour. Data: `weighted_relative_scores.csv`.

are not wins. They are a constant landing near the sample mean, and the giveaway is that `naive_historical_mean` — a constant by construction — is the best statistical model in exactly those cells.

The post-repair defect is collapse. Once goods rationing stops, the labour block overheats. Bias moves from −1.97/−0.86/−0.06/+0.19/+0.08 pp to −2.09/−2.38/−3.12/−4.39/−4.73 pp, and MASE from 7.7/6.0/5.9/5.7/5.2 to 8.1/9.3/12.2/17.3/18.8. End-of-horizon unemployment at four origins with 64 matched-seed paths is 0.76–1.11 % under the reconciliation alone and 0.12–0.50 % with the expectation change, against 4.73–5.28 % before. **This is not credible**, it is a direct consequence of the repair, and it is the diagnosis's predicted side effect.

Two separable items of future work follow, and they should not be bundled. (a) Fetching historical annual Current Population Survey series would remove the frozen-stock defect and make the target scoreable at historical origins. (b) The post-reconciliation collapse is a labour-block behavioural issue requiring its own diagnosis of the kind in §8. *(a) does not fix (b).*

Table 20: Per-target, per-horizon detail, *all-available* track. Ratios are against the VAR(1) anchor; ranks are out of the 14 scored forecast columns. Bias is the mean error (forecast − realised) in annualised percentage points. Read the last two columns together: the repair removes a chronic level bias without touching the price block.

Target	h	n	v1 RMSE	v2 RMSE	v1 ratio	v2 ratio	v1 rank	v2 rank	v1 bias	**v2 bias**
`real_gdp`	1	61	6.2592	6.0879	0.689	**0.670**	4/14	**1/14**	−1.317	**−0.025**
`real_gdp`	2	60	9.5104	6.1001	1.294	**0.830**	12/14	**1/14**	−7.277	**−0.096**
`real_gdp`	4	58	6.8762	6.2235	1.082	0.980	8/14	3/14	−2.938	**−0.143**
`real_gdp`	8	54	6.7381	6.4418	0.858	0.820	8/14	3/14	−2.146	**−0.119**
`real_gdp`	12	50	6.9637	6.6510	1.035	0.988	10/14	3/14	−2.017	**−0.202**
`gdp_deflator`	1	61	1.4288	1.4288	0.794	0.794	6/14	5/14	−0.192	−0.192
`gdp_deflator`	2	60	1.6486	1.6544	0.771	0.774	5/14	8/14	−0.293	−0.292
`gdp_deflator`	4	58	1.9398	1.9512	0.822	0.827	2/14	6/14	−0.400	−0.421
`gdp_deflator`	8	54	2.0391	2.0448	0.975	0.978	4/14	5/14	−0.525	−0.528
`gdp_deflator`	12	50	2.1093	2.1080	0.892	0.891	5/14	4/14	−0.608	−0.615
`nominal_gdp`	1	61	7.0266	6.8395	0.674	**0.656**	4/14	**1/14**	−1.509	−0.217
`nominal_gdp`	2	60	10.1803	6.7821	1.168	**0.778**	12/14	**1/14**	−7.569	−0.388
`nominal_gdp`	4	58	7.7626	7.0431	1.051	0.954	7/14	3/14	−3.338	−0.563
`nominal_gdp`	8	54	7.7071	7.3312	0.882	0.839	7/14	5/14	−2.672	−0.647
`nominal_gdp`	12	50	7.9643	7.5800	1.028	0.979	10/14	4/14	−2.625	−0.816
EFFR	1	61	0.3734	0.3734	0.563	0.563	6/14	7/14	−0.042	−0.042
EFFR	2	60	0.6921	0.6921	0.556	0.556	5/14	4/14	−0.087	−0.088
EFFR	4	58	1.2188	1.2196	0.588	0.589	3/14	4/14	−0.196	−0.199
EFFR	8	54	1.8935	1.8944	0.796	0.796	1/14	2/14	−0.500	−0.501
EFFR	12	50	2.1670	2.1646	0.806	0.805	3/14	2/14	−0.787	−0.784

Table 21: MAE ratios against the VAR(1) anchor, $h = 1/2/4/8/12$. Every real- and nominal-GDP MAE ratio is now below one at every horizon. The first pass's weak MAE column was neither a functional-choice artifact nor an intrinsic property of ensemble forecasting: it was the 2 pp bias, which costs MAE proportionally more than it costs RMSE.

Target	v1	**v2**
`real_gdp`	0.842 / **2.376** / 1.454 / 1.087 / 1.283	**0.691 / 0.734 / 0.879 / 0.790 / 0.962**
`nominal_gdp`	0.819 / **2.071** / 1.272 / 1.008 / 1.155	**0.711 / 0.725 / 0.851 / 0.804 / 0.937**
`gdp_deflator`	0.919 / 0.780 / 0.807 / 0.861 / 0.842	0.919 / 0.779 / 0.808 / 0.866 / 0.843
EFFR	0.584 / 0.627 / 0.664 / 0.762 / 0.777	0.584 / 0.627 / 0.663 / 0.762 / 0.775

Table 22: Empirical interval coverage, pooled over five horizons, $n = 283$ per target. Nominal coverage is 0.90 / 0.80 / 0.50. Computed by n-weighted aggregation of the per-horizon rows in `abm_v2_interval_coverage.csv`.

Target	v1 5–95 %	**v2 5–95 %**	v1 10–90 %	**v2 10–90 %**	v1 25–75 %	**v2 25–75 %**
`real_gdp`	0.654	**0.929**	0.580	**0.894**	0.300	0.696
`nominal_gdp`	0.661	**0.890**	0.548	**0.830**	0.332	0.608
`gdp_deflator`	0.742	0.735	0.703	0.703	0.449	0.470
EFFR	0.509	0.512	0.431	0.438	0.247	0.251
`unemployment_rate`	0.509	**0.216**	0.375	**0.127**	0.177	0.071

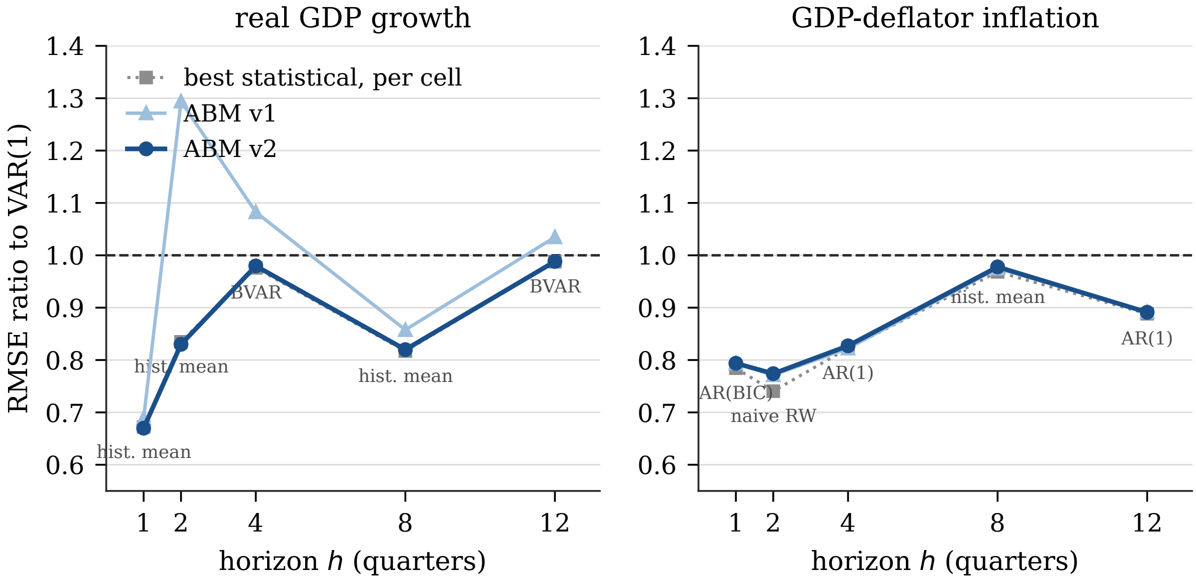

Figure 5: F2 — RMSE ratio against horizon. Left: real GDP growth — the repair moves the whole profile below the anchor and removes the $h = 2$ spike. Right: GDP-deflator inflation — the two ABM curves are indistinguishable, because the reconciliation is a quantity intervention. Grey squares mark the best statistical model *in each cell separately*, with its identity annotated; that envelope is a stiffer comparison than any single benchmark. Data: `relative_scores.csv`, all-available track.

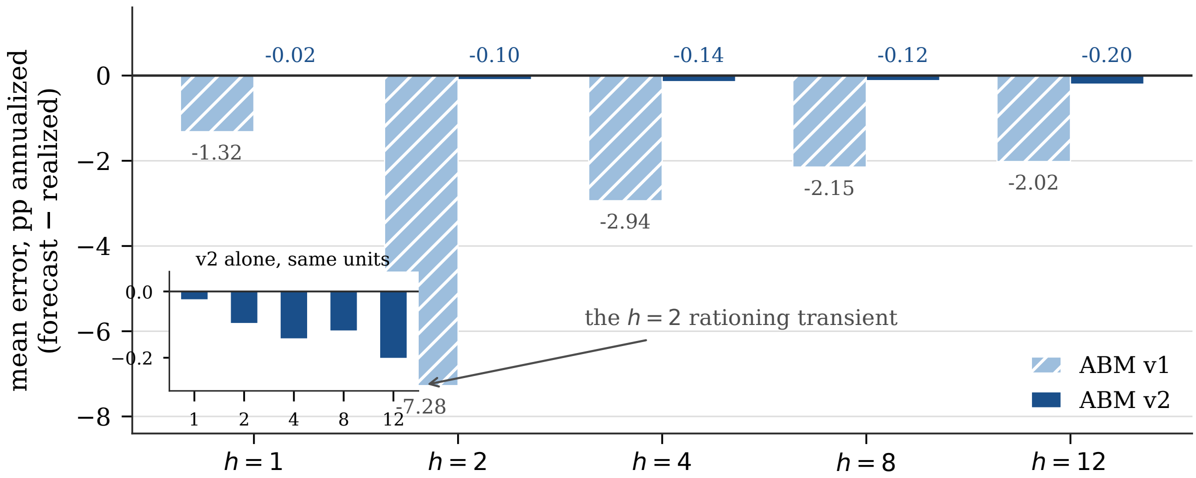

Figure 6: F3 — real-GDP mean error by horizon. The first pass under-forecasts at every horizon, with a −7.28 pp transient at $h = 2$ that the diagnosis attributes to the rationing crash rather than to the opening-row measurement basis. The repaired bars are barely visible at this scale; the inset shows them on their own axis, where the largest residual bias is −0.20 pp. Data: `score_summaries.csv`, all-available track.

Table 23: Secondary target pair {nominal GDP growth, effective federal funds rate}, weighted RMSE ratios. Weighted MAE ratios follow the same pattern: 0.717 / 0.694 / 0.735 for the repaired model against 0.971 / 0.912 / 1.268 for the first pass.

Track	v2 mean (rank)	v2 median	v1 mean (rank)	Best statistical
all-available	**0.716 (1st)**	0.716 (2nd)	0.782 (5th)	`univariate_ar_bic` 0.780
balanced h=12	**0.703 (1st)**	0.704 (2nd)	0.761 (4th)	`univariate_ar_bic` 0.770
pandemic-masked	0.739 (2nd)	0.741 (3rd)	1.124 (10th)	**`univariate_ar_p4` 0.719 (1st)**

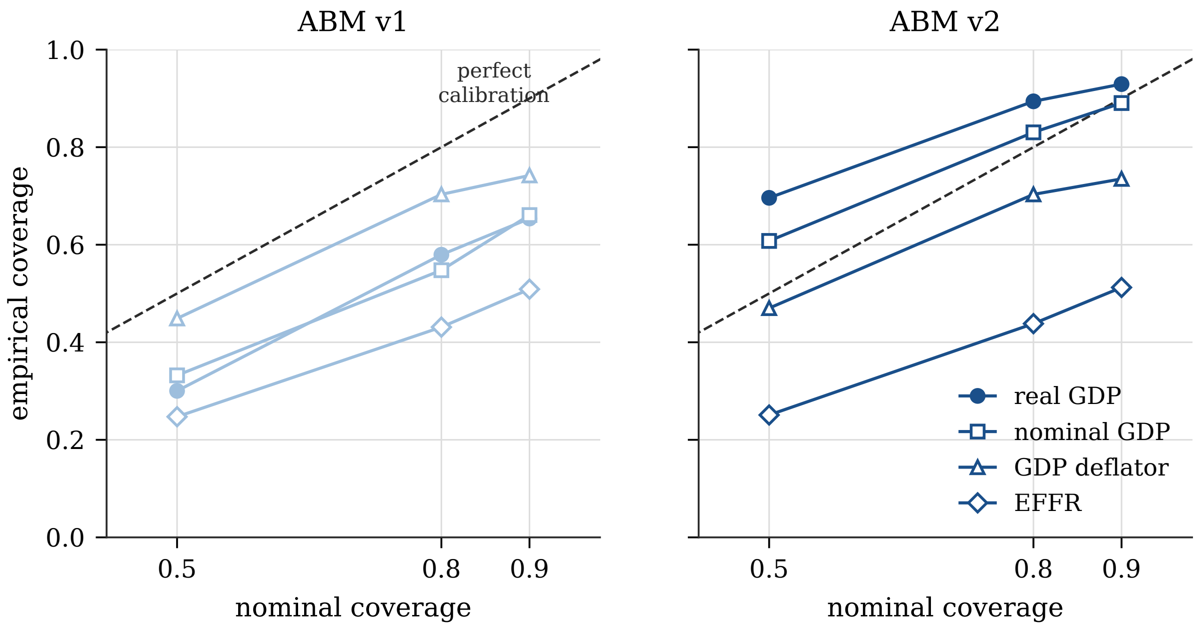

Figure 7: F4 — empirical against nominal interval coverage, pooled over five horizons, $n = 283$ per target. Points on the dashed 45° line are perfectly calibrated; below the line is over-confident. The repair lifts real and nominal GDP onto the line at the 80 % and 90 % levels while leaving the deflator and the policy rate — both under-dispersed — where they were. Data: `abm_v2_interval_coverage.csv`.

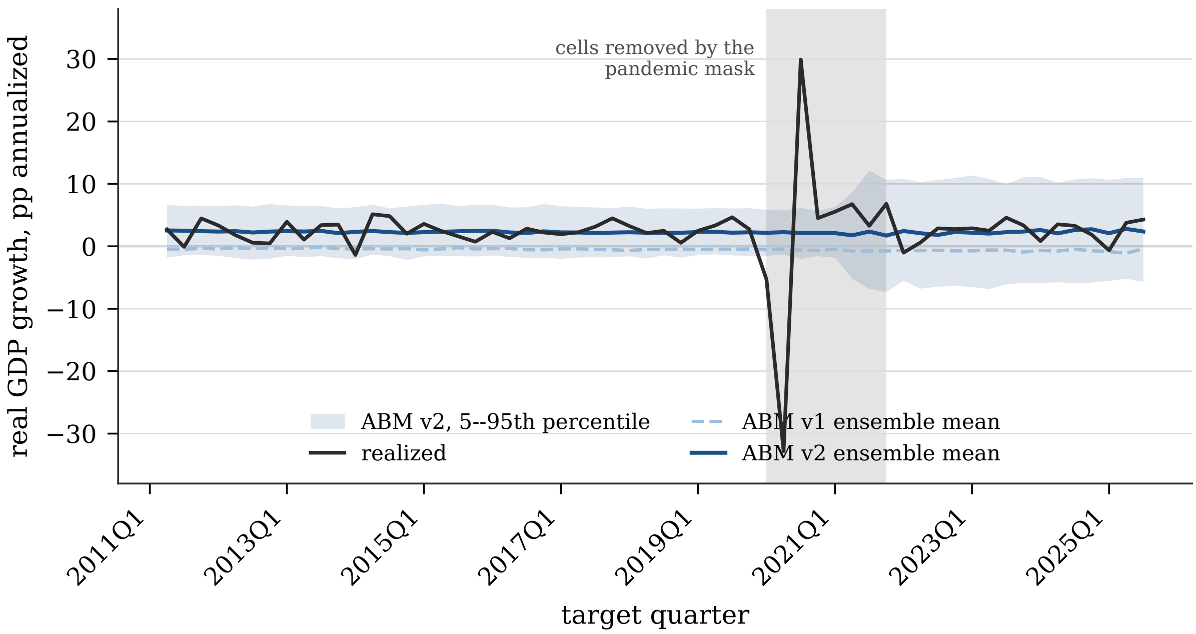

Figure 8: F9 — four-quarter-ahead real GDP growth across the whole backtest. Each point on the model curves is the ensemble functional from a different origin. The first pass sits at approximately zero growth throughout; the repaired model sits near +2.3 % and tracks the level of realised growth. Neither forecasts the pandemic quarters, which is why the pandemic-masked track exists. Only cells with a realised target quarter in the panel are shown, so the series ends at 2025Q3. Data: `abm_ensemble_summaries.csv` (both versions), `quarterly_panel.csv`.

11 The current outlook — unscored

Two origins beyond the end of the revised panel — 2025Q4 and 2026Q1 — were run on the repaired artifact with 500 paths and zero path failures. There is no realised truth for these, so **nothing in this section is scored** (`scored = false`, `realized_truth_available = false`), and the headroom caveat of §12.1 applies with more force at $h = 12$ than inside the scored window.

From the 2025Q4 origin, average real growth over $h = 1 \ldots 4$ is +2.11 % with a 5th–95th percentile band of roughly −5.5 to +10; deflator inflation runs 3.21 % at $h = 1$, easing to 2.40 % by $h = 4$ and ≈ 2.25 % by $h = 12$; the policy rate declines from 3.86 % to 3.32 %. From the 2026Q1 origin, average real growth over $h = 1 \ldots 4$ is +2.31 %, average deflator inflation +2.86 %, and the policy rate declines from 3.60 % to 3.47 % by $h = 4$ and 3.16 % by $h = 12$. Figure 9 shows the 2026Q1 fan.

Two things to note when reading Figure 9. First, the first-pass outlook's −4.9 % dive at $h = 2$ is gone: it was the transient of §7 and does not survive the balance fix. Second, path dispersion is large *by construction* — the real-GDP ensemble standard deviation is 4.6–5.1 pp, a 5th–95th band spanning ≈ 16 pp — while the Monte-Carlo standard error of the mean is ≈ 0.2 pp. The width is model dispersion, not simulation noise, and per Table 22 the real-GDP 90 % band is close to nominal coverage in backtest while the deflator and policy-rate bands are demonstrably too narrow.

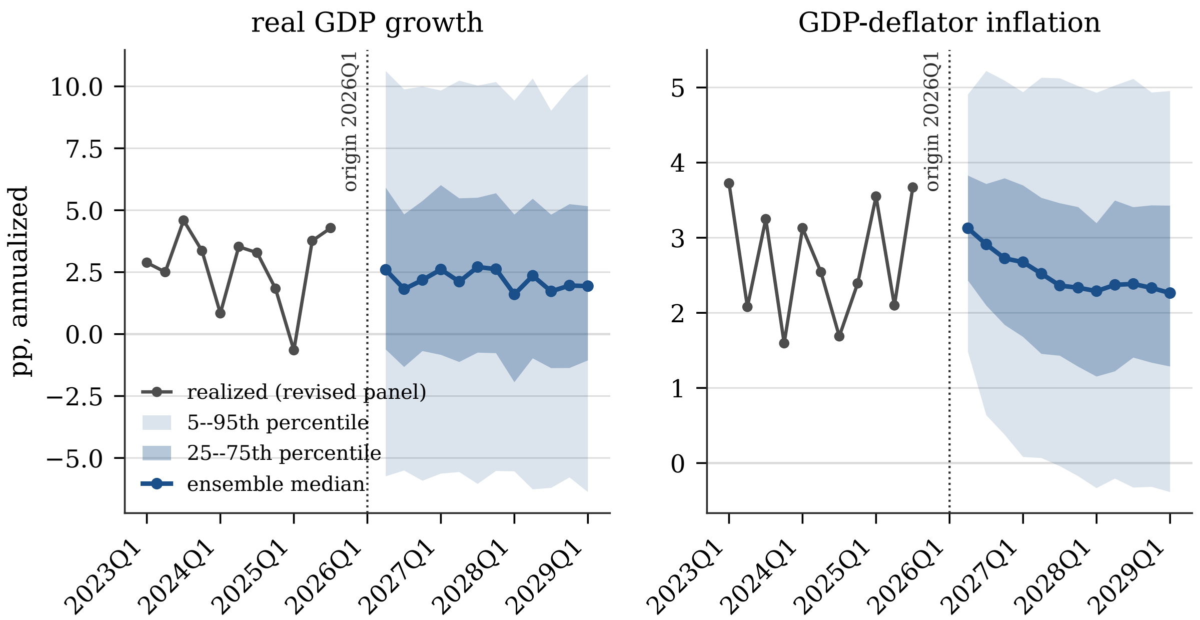

Figure 9: F6 — current outlook from the 2026Q1 origin, unscored. Realised history from 2023Q1 is prepended in grey for context; the fan shows the ensemble median with 25–75th and 5–95th percentile bands over $h = 1 \ldots 12$. The dotted line marks the origin. The deflator fan narrows to a stable $\approx 2.3\,\%$ median while the real-growth fan is essentially flat around $+2\,\%$ with wide, roughly horizon-invariant dispersion — the signature of a model whose conditional uncertainty saturates quickly. Data: `current_outlook.csv`, `quarterly_panel.csv`.

12 Limitations and open problems

Ranked by inferential risk to the results above.

12.1 Capacity-headroom growth ceiling — no capital accumulation

The most binding limitation on any forward use. Investment is replacement-only and $K_{\text{end}}/K_1 = 1.0000$ in every variant, so *all* of the repaired model's extra growth comes from draining the fixed 17.6 % opening capacity headroom implied by the hard-coded $\omega = 0.85$. Measured over 24 quarters at the 2016Q4 origin with 24 paths:

Table 24: Six-year growth decomposition by variant, annualised by year. The rejected "budgets-held" closure exhausts its headroom in about four years and decays to +0.33 %/yr, which is precisely why its smaller headline bias was not taken as evidence in its favour.

Variant	y1	y2	y3	y4	y5	y6	Utilisation y1→y6	K_{24}/K_0
v1 baseline	−1.62	−0.04	0.59	0.45	1.19	1.36	0.845 → 0.865	1.0003
budgets held (rejected)	3.33	3.60	3.53	2.71	1.39	**0.33**	0.867 → **0.986**	1.0000
v2 (λ closure)	1.17	1.06	1.56	1.47	2.86	3.33	0.856 → **0.943**	1.0000

Inside the 12-quarter scoring window this is a genuine improvement. Beyond roughly eight to ten years the repaired model must stall, because it has consumed 0.087 of its 0.144 headroom in six. A capacity-expansion term in desired investment — calibrated against the accounting identity net $I = \Delta K$, *never* against forecast RMSE — is the prerequisite for any multi-year use. **The repaired model is validated at $h \leq 12$ only, and is not a multi-year growth mechanism.**

12.2 The labour block

Two distinct failures, as set out in §10.6: a frozen 2024 opening stock, and a post-repair collapse to $\approx 1\,\%$ unemployment. Fixing the first does not fix the second, and the labour block needs its own diagnosis before unemployment is scored for either version.

12.3 Mixed-vintage structure and a single structural year

The 2024 input–output structure, firm and employee counts, tax rates and population are future information at every origin before 2024. This is bounded for flow targets and fatal for stock-initialised ones, and it is the limitation that most sharply separates this exercise from the reference paper. Separately, even setting look-ahead aside, the model has *one* technology matrix for a fifteen-year sample: real changes in U.S. production structure over 2010–2025 — the energy mix, the information-sector share, post-2020 supply-chain reconfiguration — are invisible to it by construction. This limits what the sector-level diagnostics of §8 can be asked to bear, and it is a reason to treat the reconciled 2024 table as a *fixed technology* rather than a historical one.

12.4 Absent bridges: PCE, core PCE, payroll employment

Three of the eight registered targets are unserved. The personal consumption expenditures price index and its core variant need a consumption-basket price bridge that the 68-commodity model does not expose, and the core additionally needs a food/energy exclusion the classification does not identify cleanly. Payroll employment targets an establishment count on a different universe from the model's firm employment. None should be scored until a bridge qualification of the kind done for the GDP operator is complete. The practical cost is that the model is compared on five of eight targets, and the semi-structural comparator's core-four target set cannot be matched.

12.5 Density evaluation is partial; the scale ladder was not run for the headline

§10 reports interval coverage only. Moreover all headline results are at scale 10^{-5} (130 firms). The scale sweep established that the *mean* bias is invariant from 130 to 2 708 firms, but 54 of 68 sectors hold exactly one firm at this scale, which makes cross-sectional dispersion and tail behaviour unreliable. A production run at scale $\geq 10^{-4}$ costs roughly $8\times$ the runtime and should precede any density claim.

12.6 No DSGE benchmark; the semi-structural comparator is not in the field

The reference paper compares its ABM against a Bayesian DSGE model of the Smets and Wouters [2007] family; this exercise does not. A small New-Keynesian design exists in the repository and the Federal Reserve Bank of New York's open-source DSGE model is the natural external candidate. Until one is scored on this grid, "best in the field" means *best among ten time-series benchmarks*. Adding the semi-structural Kalman comparator under the ABM's target set is the cheaper first step.

12.7 Inference

No formal inference was performed: no Diebold and Mariano [1995] test, no heteroskedasticity-and-autocorrelation-consistent adjustment, no bootstrap, and no multiple-comparison control across the $14 \times 5 \times 5 \times 3$ grid of models, targets, horizons and tracks. The rankings are descriptive.

The version contrast is the most defensible comparison in this paper because it is matched-seed on identical cells with one deliberate change; the comparisons against statistical models are not, and margins of a few per cent — such as 0.830 against `naive_historical_mean`'s 0.879 — should not be treated as established.

12.8 Residual reproducibility notes

The pipeline's baseline builder errors out and cannot regenerate its own baselines; the checked-in artifacts load and run correctly, so this does not block the work, but the provenance chain is broken at the regeneration step. The reconciliation builder is deliberately standalone and is not wired into the artifact-build entry point for that reason. Minor: the Monte-Carlo error file in the repaired run's output directory carries a stale first-pass model-family label on rows that are the repaired ensemble's own errors; the values are correct, only the family string is wrong, and no reported number depends on it. Finally, the narrative in the repository's results report states 543 RAS iterations where the committed reconciliation report states 623; the committed artifact is authoritative and Table 17 follows it.

12.9 Pointer to a vintage-clean design

A stage-3, pseudo-real-time design requires year-by-year annual history for the 37 frozen arrays — BEA input–output summary use/make tables after redefinitions; Census SUSB and BLS QCEW for firms and employees per sector; BLS CPS for population, unemployed and inactive; BEA national income and product accounts for tax and benefit aggregates, and BEA fixed-assets tables for assets, dwellings and capital consumption [U.S. Bureau of Economic Analysis, 2025b] — plus a real-time truth layer of the Croushore and Stark [2001] kind, for which a Philadelphia Fed real-time data set capture already exists on disk. Quarterly series need no fetch; they run from 1996Q4. This is a data-acquisition programme, not a modelling one, and it is the only route to a promotion claim. A realistic first cut restricts origins to 2014 and later, where the annual coverage is densest.

13 Conclusion

An agent-based macroeconomic model of the United States was calibrated from public source data, scored against ten statistical benchmarks on a matched grid of 61 quarterly origins and five horizons with 500 free-running paths per origin, found to carry a large and systematic real-growth bias, diagnosed by matched-seed experiment to two specific defects, repaired by restoring an accounting identity and correcting a misspecified estimator — neither repair *selected* using forecast errors — and then re-scored on identical cells with identical seeds. The exercise remains explicitly mixed-vintage: 2024 structure and current-vintage data are used at historical origins. That arc is the result. The rank improvement is a consequence of it, not the object of it.

Three findings are worth carrying away.

First, the defect was in the accounting, not the behaviour. A supply–use system that balances in its published valuation basis need not balance after a valuation bridge and an industry-to-commodity projection, and an agent-based model with a fixed capacity envelope and a one-sided production constraint has no mechanism that can absorb the difference. The failure mode was not a badly chosen parameter; it was an identity that had quietly stopped holding. This is a class of failure that structural simulators are especially exposed to and that a reduced-form model, having no accounting identities to violate, cannot exhibit.

Second, the diagnosis made a falsifiable out-of-sample prediction, and it held. The mechanism implied a real quantity-rationing defect and therefore predicted that the price block would be untouched by the repair. The GDP deflator's $h = 1$ cell is bit-identical between the two versions, and its ratios differ in the third decimal at every other horizon, while real-GDP bias falls by two orders of magnitude. A diagnosis that predicts what will *not* change is more informative than one that only predicts what will.

Third, the improvement is bounded by a defect that was not repaired. All of the repaired model's growth drains a fixed 17.6 % capacity headroom, because investment is replacement-only and capital never accumulates. Within the twelve-quarter scoring window this is a genuine improvement; beyond roughly eight to ten years it is not a growth mechanism at all. Reporting the rank without this caveat would misrepresent what has been established.

What has been established is that a specific, located, repairable accounting defect explained a 2-percentage-point forecast bias, and that after repair the model attains the lowest weighted RMSE ratio in a fourteen-column field on the headline target pair in all three sample tracks. What has not been established is real-time forecast skill, validity beyond twelve quarters, anything about the labour block, or statistical separation from the models ranked immediately below. The route from here is a data-acquisition programme for vintage-correct structural inputs, a capacity expansion term calibrated against the investment identity, and a labour-block diagnosis of the kind performed here for the goods market.

A Reproducibility

A.1 Commands

All commands run from the repository root. Both environment variables are hard requirements on the benchmark path: the runner raises an error unless the Julia and BLAS thread counts are both 1.

```
# 1.  build the reconciled calibration artifact (∼40 s)
julia -project=scripts/us \
    scripts/us/calibration/reconcile_commodity_balance.jl \
    -mode=final_demand_scaled -expectations=rw_drift
# 2.  the two ensembles, 61 origins × 500 paths each
JULIA_NUM_THREADS=1 OPENBLAS_NUM_THREADS=1 julia -project=scripts/us \
    scripts/us/forecasting/diagnostics/abm_revised_comparison/\
    run_revised_data_abm_comparison.jl \
    output/us_forecasting/abm_v2_comparison/v1_headline 500 headline
    ...  v2_headline 500 headline_v2
# 3.  joint scoring:  both ABM columns on identical common cells
    ...  v2_headline 500 headline_v2 -also-score=.../v1_headline
# 4.  current outlook (unscored)
    ...  output/us_forecasting/abm_v2_comparison_outlook 500 outlook_v2
# 5.  figures for this paper
python3 paper/figures_src/make_figures.py
# 6.  this paper
cd paper && latexmk -pdf US_ABM_Forecasting_Paper.tex
```

Step 2 resumes from its own ensemble cache, so an interrupted run continues rather than restarting; step 3 is a re-score off the cache (about 15 s) rather than a re-simulation. Every figure in this paper is regenerated by step 5 directly from the committed CSVs listed in §A.2; no figure contains a hand-entered number, and the generator pins `SOURCE_DATE_EPOCH` so that a re-run reproduces each figure byte for byte.

Path (under the repository root)	Contents
`output/us_forecasting/abm_v2_comparison/v2_headline/`	the canonical scored result: both ABM columns, 14 forecast columns, 500 paths — `manifest.toml`, `score_summaries.csv`, `relative_scores.csv`, `weighted_relative_scores.csv`, `monte_carlo_errors.csv`, `abm_v2_interval_coverage.csv`, `abm_ensemble_summaries.csv`, `abm_origin_diagnostics.csv`, `failures.csv`
`output/us_forecasting/abm_v2_comparison/v1_headline/`	first-pass ensemble cache, re-run under the patched tree (gate G4)
`output/us_forecasting/abm_v2_comparison_outlook/`	2025Q4 and 2026Q1 origins on the reconciled artifact, unscored
`output/us_forecasting/commodity_balance_reconciliation/`	`reconciliation_report_rho1.txt`, `reconciliation_by_commodity_rho1.csv`
`scripts/us/forecasting/diagnostics/revised_data/fixtures/quarterly_panel.csv`	the scoring panel: 101 quarters, 2000Q3–2025Q3
`paper/figures_src/make_figures.py`	the figure generator for this paper

A.2 Artifact paths

A.3 Commit hashes and identity seals

Commit	Content
`a55d9ed`	checkpoint before the ABM campaign; the base all gates compare against
`9430a4a`	first-pass comparison: ABM against statistical benchmarks
`cd22674`	random-walk-drift expectations patch (flag-gated, default off)
`b82680e`	RAS reconciliation of the opening commodity balance + reconciled artifact
`aadde2b`	repaired-model comparison run
`107a9ef`	merge of the repaired line
`461cdc3`	results report, from which this paper's narrative is drawn

Table 25 lists the fields sealed in every repaired-run manifest. For contrast, the first-pass calibration object is `data/us/calibration/US_2024_calibration_object.jld2` with SHA-256 `4cce5d62...05e2e136`.

The λ semantics recorded verbatim in each manifest read: *"explicit accounting choice: the four final-demand aggregates C, G, I and X (and capital_consumption and gross_capitalformation_dwellings, which set the investment budget) are scaled by lambda so the artifact's expenditure aggregates match its production account and the opening commodity balance clears exactly. lambda is fixed by the accounting identity alone and was not chosen with reference to any forecast error."*

A.4 Seed policy

- Simulation seeds are hash((`:abm_revised_v1`, origin, path)) mod 2^{30}: domain-separated by origin and path, so no two cells share a stream.
- The repaired model draws the *identical* seed stream as the first pass, which makes the contrast matched-seed. The expectations patch consumes exactly one Normal variate on either branch to preserve this.
- The seed is set *immediately before* model construction, because the initial microstate is drawn from the global generator; seeding after construction yields a random initial state and is a silent error. A fresh model is constructed per path; the library's built-in ensemble

Manifest field	Value
calibration_object_path	data/us/calibration/US_2024_calibration_object_ reconciled.jld2
calibration_object_sha256	57e23f4a...e514760
commodity_balance_reconciled	true
reconciliation_mode	final_demand_scaled
reconciliation_rho	1.0
reconciliation_lambda	0.9834601934561227
growth_expectation_specification	rw_drift
monte_carlo_paths / burn_in_quarters / simulated_quarters	500 / 0 / 12
model_count	14
panel_sha256	f7bb26a4...bff851bbe
julia_version / julia_threads / blas_threads	1.10.3 / 1 / 1

Table 25: Identity seals recorded by the runner in the repaired run's manifest.

runner is not used, because it deep-copies one already-constructed model and would give every path the same initial microstate.

- Diagnosis experiments use a separate domain-separated seed family, identical across every variant.
- Failed paths are counted and left visible, never resampled. The count is 0 in every run.

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